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Review on photovoltaic/thermal hybrid solar collectors: Classifications, applications and new systems

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ABSTRACT

The continuous aggravation of environmental pollutions and the rapid increase in the conventional sources of energy price advocated scientists to search for solutions such as heat recovery and continual sources of energy. One of the most distinguished types of renewable energy is solar energy which is neat, free and environmentally friendly renewable energy. Indeed, there are several solar energy systems that need to be studied. At the same time, combining solar energy systems is a trend these days which aims to optimize the benefits from solar radiations. Photovoltaic/Thermal (PVT) hybrid solar system is obtained by combining solar thermal collectors and solar photovoltaics to enable a simultaneous generation of electricity and production of heat. The target of this paper is to proffer a review on PVT hybrid solar collectors which comprises the history of PVT hybrid solar systems, main concept, benefits and classifications according to various manners in addition to integration of thermoelectric generators with PVT system.

1. Introduction

The stability and firmness provenience of human life is energy. It is tremendously needed in progress and development for a better life. However, the increase in energy demand with simultaneous aggravation of environmental problems and depletion of conventional energy sources contribute to the search for clean and sustainable sources of energy. Renewable sources of energy (Afrand et al., 2019; Ananno et al., 2020; Carvalho et al., 2019; Dong et al., 2020; Fernández-González et al., 2019a, 2019b; Guo et al., 2020; Herez et al., 2017a; Jurasz et al., 2020; Shah and Ali, 2019a; Valančius and Mikučionienė, 2020; Yahya et al., 2018) are abundantly existing and capable to provide energy in a clean way. Solar Energy (SE) (El Hage et al., 2018; Elsheikh et al., 2019; Herez et al., 2018, 2017b, 2016; Khaled et al., 2018; Molaei, 2020; Naves et al., 2019) is a type of renewable energy that can be exploited using Solar Thermal Collectors (STC) or solar PhotoVoltaic (PV).

Solar collector absorbs SE and transfers it into heat to be harnessed for various aims including heating water or/and space, cooling space and/or refrigeration, drying, cooking, etc. Solar PV (Abdul-Wahab et al., 2019; Dhoke et al., 2019; Kiyaninia et al., 2019; Lindsay et al., 2020; Olaleru et al., 2020; Paul Ayeng'o et al., 2019; Rajvikram and Sivasankar, 2019; Robledo et al., 2019; Trindade and Cordeiro, 2019; Yang et al., 2020) is a technology which gathers solar radiations to transform it into electrical energy. Indeed, the efficiency of electricity conversion by solar PV system from solar energy is about 15 to 18%, whereas the remaining part (82–85%) is either absorbed by PV cell and thus heating it or reflected to the environment (Browne et al., 2016). Furthermore, it is approximated that the increase in the module temperature by one degree leads to mitigation in the efficiency of PV Module (PVM) by 0.4 to 0.65% (Brahim and Jemni, 2017). Hence, PV systems captivated the efforts of scientists in order to enhance its efficiency.

Nevertheless, there is a great receptivity to amalgamate solar energy systems to optimize the avail of SE. Hence, PhotoVoltaic/Thermal (PVT) hybrid solar collector was suggested as a solution for promoting the PV efficiency and the benefit of solar radiation. It is incorporation of solar PV with the STC that serves in the simultaneous generation of electricity and heat with half the area needed and little extra cost. Numerous investigations are conducted since the development of solar

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hybrid systems (Chow, 2010; Gaur et al., 2017; Greppi and Fabbri, 2018; Ju et al., 2017; Qiu et al., 2015; Sahota and Tiwari, 2017; Sathe and Dhoble, 2017; Shah and Ali, 2019b).

(Jia et al., 2019) proffered a comprehensive review of the progress and application of PVT systems. The review comprises the classification of PVT collectors and a description of suitable applications and conditions for various types of PVT systems.

(Yazdanifard and Ameri, 2018) had reviewed researches that were performed during the last decade to assess the exergetic performance of several kinds of PVT hybrid systems. The review concentrates on describing the performance of water based, air based and thermoelectric based PVT systems. Also, a comparison of the exergy analysis between distinct PVT systems of various studies is performed.

(Abdelrazik et al., 2018) presented a comprehensive review including the latest works that investigated the design and cooling parameters that are employed to construct a hybrid PVT system with high performance.

(Sultan and Ervina Efzan, 2018) conducted a review that exposes the recent literature on PVT technology by describing its types, fabrication, design, simulation and experimental assessment of its performance. Also, the authors reviewed the applications of PVT technology such as water desalination, solar still, PVT heat pump, solar cooling, and solar greenhouse.

(Sahota and Tiwari, 2017) performed a review to assess and compare the economic and energy performance of several photovoltaic hybrid systems which are: photovoltaic thermal compound parabolic concentrator (PVT-CPC), photovoltaic thermal flat plate collector (PVT-FPC), PVT-FPC solar still and greenhouse dryer PVT; they concluded that integrating photovoltaic in solar systems is beneficial and more economical for the rural sector.

(George et al., 2019) presented a review on the concentrated PVT systems including the design components, medium of heat transfer and the latest applications like heating water, dyeing tissue and organic Rankine cycle as well as the economic side.

(Kasaeian et al., 2018) performed a review which comprises the literature of integrated solar collectors and photovoltaic systems for the simultaneous heat and power generation. The review included solar PVT systems, concentrated PVT systems with several combinations and applications.

(Jonas et al., 2019) established a novel PVT model to compare the impact of several factors on the system effectiveness. The model was simulated and validated using TRNSYS software.

(Huide et al., 2017) developed simulation models on solar photovoltaic, solar thermal and solar hybrid PVT systems. The authors validated the results of models simulation by performing experiments. The results revealed that for the urban residences where space is limited, the hybrid PVT system is the best choice to mitigate energy consumption.

Fine et al. (2019) demonstrated a novel methodology to assess the thermal performance of hybrid solar panel by utilizing only the supplied data by the industrialist. The authors computed the thermal performance by using the second order efficiency model and the electrical efficiency by using the temperature dependent electrical properties. They also utilized a solar simulator to carry out an indoor testing.

In this context, the main purpose of the current work is to present a detailed review on PVT hybrid solar collectors. The background of the PVT system is presented, which mainly includes the history, concept and advantages of the system. Also, the classifications of such systems are exposed depending on different methods. Besides, the review comprises the main applications that are involved in the studied system. Moreover, new systems are reviewed such as adding Thermoelectric Generators (TEGs) to the PVT system. The new contribution of this work resides in the following points:

(1) Presenting a new classification of PVT systems;
(2) Describing the applications using PVT systems;
(3) Presentation of PVT systems using TEGs (PVT-TE) with their types and operations;
(4) Summary of numerous works conducted on PVT systems regarding each classification, application and type of PVT-TE systems;
(5) Description of the most commonly used systems according to the classifications, applications and PVT-TE system.

Although this work may have partially similar points with those treated in other review papers, it combines all information on PVT systems and presents them in a categorized manner in order to constitute a valuable guide for studying PVT systems.

2. Background on PVT hybrid systems

The idea of PVT system was discovered during the mid-1970s where several studies had been conducted (Joshi and Dhoble, 2018). In 1976, Wolf (1976) investigated the liquid type PVT solar collector by monitoring the collector performance for single-family residence over a full year. In 1979, Florschuetz (1979) modified the original parameters of "Hottel Whillier model" which is thermal modeling for flat plate collector and studied the electrical and thermal performance of PVT collectors. Despite the fact that the concept of PVT solar collectors had been established for more than four decades, it is not totally commercialized yet. Several types of research had been performed recently to optimize the PVT collectors performance and minimize its cost (Fine et al., 2019; Khelifa et al., 2015; Magrassi et al., 2019; Yazdanifard and Ameri, 2018). Nowadays, PVT systems have become an appealing invention that is environment friendly and mitigates dependence on conventional sources of energy.

A PVT collector is a hybrid system which is obtained by merging solar PV cell with a STC to provide electricity and heat simultaneously. Solar PV converts the absorbed SE into electricity, whereas the STC converts it into heat. Such a system maximizes the avail of SE; so, it is more efficient and can produce higher amount of energy per unit area in comparison to solar PV or solar thermal systems alone. Fig. 1 shows a schematic of the concept of PVT hybrid solar collector.

PVT hybrid solar collector was established mainly to optimize the SE exploitation. The utilized region by PVT is greater than that used by traditional PV or thermal collectors. To clarify, with the same area, PVT can generate heat and electricity more than that produced by conventional PV or thermal collectors. This explains the fact that PVT has greater efficiency than both systems when operating separately. Also, it is suitable for places where there is a limited surface area. Besides, PVT

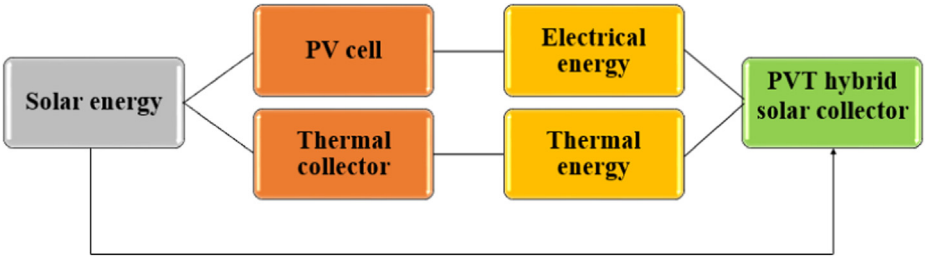


Fig. 1. Concept of PVT solar collector.

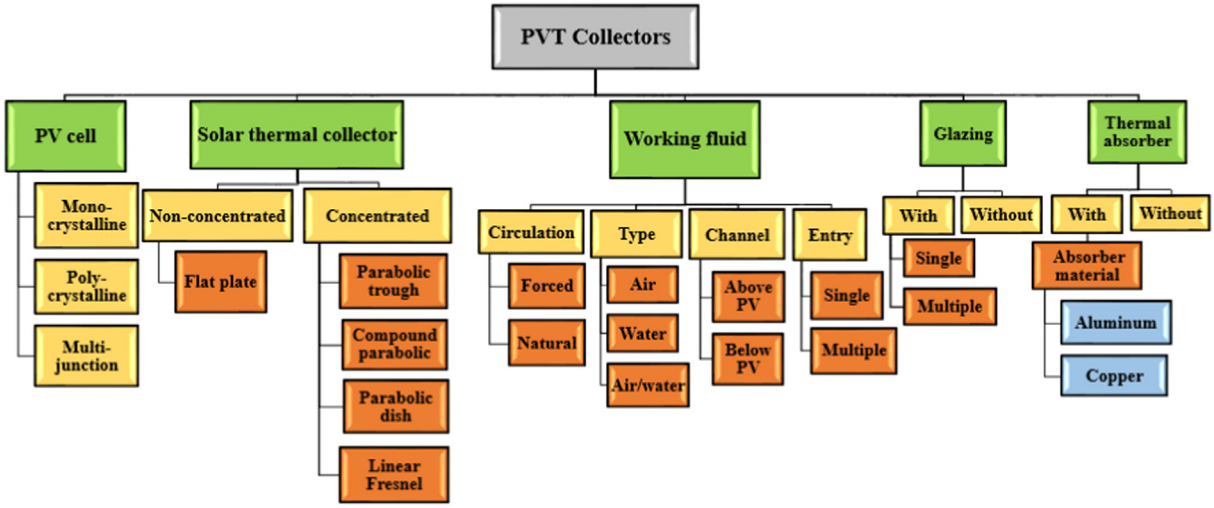


Fig. 2. Classification of PVT solar collectors.

system combines two systems with each other, and this leads to a reduction in the installation cost. Furthermore, the payback period of PVT has been obviously less than both systems since it exploits more energy.

3. Classifications of PVT

PVT hybrid solar systems can be categorized based on various characteristics. Several researchers carried out studies including the classification of PVT (Abdullah et al., 2018; Diwania et al., 2019; Evangelisti et al., 2019; Joshi and Dhoble, 2018; Lamnatou and Chemisana, 2017; Riffat and Cuce, 2011). In this work, sorting PVT is demonstrated by taking into account the type of PV cell, thermal collector, working fluid, glazing and thermal absorber (Babu and Ponnambalam, 2017; Joshi and Dhoble, 2018). Fig. 2 presents a wide classification of PVT.

3.1. PV cell

PVT collectors can be categorized regarding the type of PV cell. The most commonly used PV cells are mono-crystalline silicon (mono c-Si), poly-crystalline silicon (poly c-Si) or multi-junction. Mono c-Si and poly c-Si belong to the first generation of PV technologies while multi-function thin-film silicon belongs to the second PV generation. Mono-crystalline and poly-crystalline generate the same power when using same area; however, mono-crystalline has higher efficiency (about 20%) (Cappelletti et al., 2016) than poly-crystalline (15%) due to the grains existing in the poly-crystalline (Tao and Yu, 2015). Nevertheless, mono-crystalline silicon is more expensive than poly-crystalline due to the complex formation of a single crystal. Pang et al. (2019) experimentally demonstrated the performance of the poly-crystalline silicon and PVT module. A steady state thermal modeling was conducted to simulate the pressure, temperature, and velocity along the PVT module. The performances of PV and PVT module were compared. It was observed that the electricity generated by PVT system augmented by 3.25%.

A multi-junction solar cell is formed of more than one p-n junction. It mainly utilizes three semiconductors: germanium (Ge), gallium indium phosphide (GaInP) and indium gallium arsenide (InGaAs). It is more efficient than silicon cells (around 40%) and less sensitive to temperature variation (Mittelman et al., 2007; Nishioka et al., 2006; Renno and Petito, 2016). Nishioka et al. (2006) demonstrated the influence of temperature and concentration ratio on nGaP/InGaAs/Ge triple junction electrical performance. It was obtained that the PV cells efficiency mitigates with temperature rise and augments with concentration ratio increase because of the rise in the open circuit voltage. Renno et al. (2018) considered a specific structure of a CPV system and performed experimental analysis on the system with degraded triple junction InGaP/GaAs/Ge solar cells. The degradation of the PV cell is caused by the excessive augmentation in the light intensity which causes its overheating. The results revealed that the degradation of the cells reduces the power conversion efficiency of the CPV system by 50% and the electric power by 30%.

Table 1 illustrates the latest researches conducted on PVT hybrid systems according to the type of PV cell used.

Analysis of Table 1 yields to the following concluding points:

- Crystalline silicon and InGaP/GaAs/Ge triple junction solar cells are broadly used in PVT systems in comparison to other PV cells.
- InGaP/GaAs/Ge triple junction PV cell yields to higher electrical efficiency of PVT system compared to crystalline silicon. This is due to the fact that it has higher electrical efficiency than crystalline silicon.
- PVT system thermal efficiency is approximately similar when using crystalline silicon or InGaP/GaAs/Ge triple junction PV cell.

3.2. Solar thermal collector

PVT hybrid solar systems are sorted according to the type of STC combined with PV cell whether it is a non-concentrated collector like a Flat Plate Collector (FPC) (Michael et al., 2015; Moss et al., 2018; Sakellariou and Axaopoulos, 2018; Zhang et al., 2019) or concentrated like a parabolic trough collector (PTC) (Karathanassis et al., 2017; Kunnemeyer et al., 2014; Manokar et al., 2016; Mittelman et al., 2007; Widyolar et al., 2017), Compound Parabolic Concentrator (CPC) (Atheaya et al., 2016a; Li et al., 2019; Saini et al., 2018; Singh et al., 2016; Tiwari et al., 2018; Tripathi et al., 2016; Tripathi and Tiwari, 2016), Parabolic Dish Collector (PDC) (Singh and Reddy, 2020) and Linear Fresnel Reflector (LFR) (Pham et al., 2018; Shanks et al., 2018; Sharshir et al., 2016).

FPC is made up of flat absorber plate, transparent cover, working fluid, tubes, insulation and protective case. It is used for applications that require low temperature (30–80 °C) (Herrando et al., 2014). It is widely utilized in Solar Space Heating (SSH) and Solar Water Heating (SWH) systems. Das et al. (2018) presented a detailed review on the PVT systems coupled to a flat plate collector. Their review includes an environmental and economic assessment of PVT systems that were established by several researchers. Also, it comprises listing thermodynamic ways for assessing efficiency of the system such as energy and exergy analysis. Guarracino et al. (2019) studied the performance characteristics of a non-concentrating flat plate PVT collector which is

Table 1
Conducted researches on hybrid systems according to the type of PV cell.

St. Nb	Reference (Author/s, year)	PV cell	Operating conditions	Efficiency Thermal	Electrical	Total	Type of study
	Ibrahim et al. (2009a)	Poly-crystalline silicon	• Solar radiation of 800 W/ m^2 • Ambient temperature of 25 °C	45%	10%	55%	• Experimental
	Bernardo et al. (2011)	Mono-crystalline silicon	• Incident beam radiation of 997 W/ m2 • Water inlet temperature of 28 °C	45%	6.4%	–	• Experimental
	Chemisana et al. (2011)	Crystalline silicon	–	65%	–	–	• Experimental • Theoretical
	Sonneveld et al. (2011)	Mono-crystalline silicon	• Solar radiation of 630 W/ m2 and 792 W/ m2	56%	11%	67%	• Experimental
	Dupeyrat et al. (2011)	Mono-crystalline silicon	• Total radiation of 960 W/ m2 • Ambient temperature varied between 30 and 33 °C • Wind velocity of 3 m/s	79%	8.8%	86.8%	• Theoretical • Experimental
	Calise and Vanoli (2012)	InGaP/InGaAs/Ge	• Inlet water temperature of 70 °C • Ambient and sky temperature of 25 °C • Wind velocity of 5 m/s • Total and beam radiation of 1000 W/ m2 and 800 W/ m2 respectively	56%-63%	20%–23%	77%-83%	• Theoretical
	Buonomano et al. (2013)	InGaP/InGaAs/Ge triple junction	• Inlet fluid temperature of 150 °C • Ambient and sky temperature of 25 °C • Wind velocity of 5 m/s • Total and beam radiation of 1000 W/ m2 and 800 W/ m2 respectively	27%-58%	18%-24%	45%-82%	• Theoretical
	Renno and Petito (2013)	InGaP/InGaAs/Ge triple junction	–	67%-70%	19%-21%	86%-91%	• Theoretical
	Del Col et al. (2014)	InGaP/InGaAs/Ge triple junction	• Inlet water temperature ranges between 20 and 80 °C • Beam radiation of 1000 W/ m2	41.6%	23.1%	64.7%	• Theoretical • Experimental
	Hussain and Lee (2015)	GuAs and Ge	• Solar radiation of 900 W/ m2 • Fluid inlet temperature of 25 °C	68%	–	–	• Theoretical • Experimental
	Xu et al. (2015)	InGaP/InGaAs/Ge triple junction	• Ambient temperature between 15 and 17 °C • Beam radiation of 700 W/ m2	54%	28%	84%	• Experimental
	Yew et al. (2015)	Triple junction	• Solar irradiance of 1000 W/ m2 • Default temperature of 25 °C	34.1%-38.7%	20%-25%	–	• Theoretical
	Mi et al. (2016)	GaInP/GaInAs/Ge triple junction	• Solar irradiance of 900 W/ m2 • Ambient temperature of 25 °C	–	–	–	• Experimental • Theoretical
	Cui et al. (2016)	GaInP/InGaAs/G, c-Si, CIGS and GaAs	• Ambient temperature of 27 °C • Water inlet temperature of 20 °C • PV cell temperature between 21 and 35 °C • Total solar irradiance of 1000 W/m2	–	–	25%-27%	• Theoretical
	Mojiri et al. (2016)	Silicon	–	50%	3.8%	–	• Theoretical • Experimental
	Widyolar et al. (2017)	GaAs	• Beam radiation of 700 W/m2 • Fluid inlet temperature between 100 and 500 °C	37%	8%	–	• Theoretical • Experimental
	Ceylan et al. (2016)	Silicon	• Ambient temperature between 20 and 40 °C • Total radiation of 1000 W/m2		11%	–	• Theoretical • Experimental
	Gürel (2016)	Silicon	• Total radiation of 1000 W/m2 • PV cell temperature of 25 °C	–	–	11%-16%	• Experimental
	Xu et al. (2016)	InGaP/InGaAs/Ge	• Ambient temperature of 25 °C • Wind velocity of 4 m/s • Beam radiation of 1000 W/m2	60%	28%	–	• Theoretical
	Karimi et al. (2017)	Mono-crystalline	• Solar radiation of 920 W/m2 • PV cell temperature of 20 °C	43.6%	14.5%	53%	• Experimental
	Burhan et al. (2017)	Mono-crystalline, poly-crystallineand thin-films CIS	–	–	22%	23%	• Theoretical
	Srivastava and Reddy (2017)	InGaP/GaAs/Ge	• Fluid inlet temperature and ambient temperature of 27 °C • Beam radiation of 650 W/m2	69.68%	18.91%	85.82%	• Theoretical • Experimental
	Yazdanifard et al. (2017)	GaAs	• Fluid inlet temperature and ambient temperature of 25 °C • Solar radiation of 700 W/m2 • Wind velocity of 1.5 m/s	52.58%-56.37%	7.12%-7.94%	61%-63.5%	• Theoretical
	Proell et al. (2017)	Crystalline silicon	• Ambient temperature between 17 and 29 °C • PV cell temperature of 22.9 °C • Total radiation of 1000 W/m2	34%	9%	–	• Experimental
	Chong et al. (2017a)	Triple junction	• Solar irradiance of 1000 W/m2	–	–		• Experimental

(*continued on next page*)

Table 1 (*continued*)

St. Nb	Reference (Author/s, year)	PV cell	Operating conditions	Efficiency Thermal	Electrical	Total	Type of study
						16.1%-17.4%	• Theoretical
	Schmitz et al. (2017)	InGaP/GaAs/Ge	• Fluid inlet temperature of 85 °C	–	26.6%	–	• Theoretical
	Lu et al. (2018)	Crystalline silicon	• Incident radiation intensity of 280 W/m2 • Ambient temperature of 23.9 °C	–	14.8%	–	• Theoretical • Experimental
	Sakellariou and Axaopoulos (2018)	Poly-crystallinesilicon	• Wind speed between 0 and 6 m/s • Ambient temperature between 24 and 28 °C • Incident solar irradiance between 200 and 900 W/m2	–	–	–	• Theoretical • Experimental
	Aoul et al. (2018)	Mono-crystalline	• Ambient temperature between 21 and 45 °C • Wind speed between 0 and 5 m/s	24%-36%	20%-24%	30%-47%	• Experimental
	Jaaz et al. (2018)	Poly-crystalline silicon	• Ambient temperature of 33 °C • Solar radiation of 800 W/m2	81%	13.75%	91%	• Experimental
	Soltani et al. (2018)	Crystalline silicon	• Ambient temperature of 36.2 °C • Water inlet temperature of 28 °C • Solar radiation of 900 W/m2	57%	–	–	• Theoretical
	Adam et al. (2019)	Silicon	• Ambient temperature of 25 °C • fluid inlet temperature of 20 °C • Direct normal irradiance of 1000 W/m2	55%-70%	5%-18%	66%-85%	• Theoretical
	Bellos and Tzivanidis (2019)	Mono-crystalline silicon	• Solar irradiation of 1000 W/m2 • Reference temperature of 25 °C	46.84%	6.6%	–	• Theoretical
	El-samie et al. (2019)	Poly-crystalline silicon	• Reference temperature of 25.15 °C	48%-50.6%	6.9%-7.3%	80%-89%	• Theoretical
	Shittu et al. (2019)	Poly-crystalline silicon	• Ambient temperature between 15 and 55 °C • Wind speed between 0 and 7 m/s	–	–	16%-17%	• Theoretical
	Yu et al. (2019)	Poly-crystalline silicon	• Ambient temperature between 21.6 and 23.6 °C • Wind speed between 2.2 and 3 m/s • Solar irradiance between 880 and 940 W/m2 • PV cell temperature between 20 and 45 °C	24.2%-25.2%	11.8%	36%-37%	• Theoretical • Experimental
	Zhang et al. (2019)	Poly-crystalline silicon	• Ambient temperature of 18 °C • Wind speed of 0.1 m/s • Solar irradiance of 900 W/m2 • Inlet fluid temperature of 14.5 °C	56.7%	11.8%	66.4%	• Theoretical
	Valizadeh et al. (2019)	InGaP/GaAs/Ge	• Ambient temperature of 25 °C • Sky temperature of 19 °C • Wind speed of 5 m/s • Beam radiation of 800 W/m2 • Total radiation of 1000 W/m2 • Inlet fluid temperature of 70 °C	59%-67%	15%–22%	–	• Theoretical
	Bamroongkhan et al. (2019)	Mono-crystalline silicon	• Solar intensity between 300 and 826 W/m2 • Ambient temperature around 30 °C	–	19.65%	–	• Theoretical
	Wang et al. (2019)	Mono-crystalline silicon	• Ambient temperature between 30 and 33 °C • Wind speed of 3 m/s • Solar radiation of 960 W/m2	79%	8.7%	88%	• Theoretical

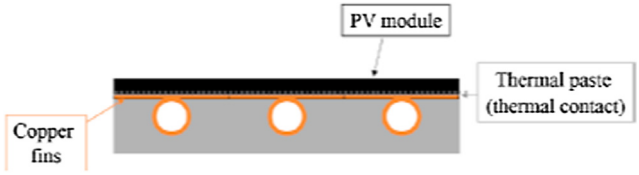


Fig. 3. Flat plate PVT collector (Guarracino et al., 2019).

presented in Fig. 3. Experimental tests were carried out on four PVT collectors: commercial PVT module, unglazed PVT module with 65% PV covering factor and glazed PVT modules with 65% and 100% PV covering factor. It was obtained that the PVT collector thermal efficiency alleviates by 10% when the electricity conversion is relatively maximized compared to open-circuit mode. Also, the collector thermal performance deteriorates when there is weak thermal contact of the thermal absorber with the PV layer. Besides, it was shown that the glazing enhances thermal efficiency and mitigates electrical efficiency.

Moreno et al. (2017) studied an unglazed flat plate PVT system by analyzing its performance at distinct operating conditions. A finite element modeling was carried out, simulated and compared with an existing theoretical mathematical model. The results revealed that the PVT system overall performance is higher than that of the thermal system and it is significant for places with limited areas even though the outlet temperature is relatively moderate. Lämmle et al. (2017) developed, validated and simulated, by using TRNSYS, an empirical model to evaluate the performance of glazed, unglazed and hybrid PVT-FPCs of low emissivity coatings. It was detected that the glazed PVT collector attains the highest global yields and the best energetic exploitation of the collector area. In addition, it was observed that with the same thermal output, the electrical output of these collectors is greater than that of PVM and FPC operating separately by about 3 times. Rashmi

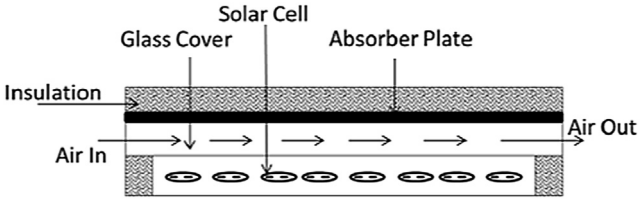


Fig. 4. Flat plate PVT collector (Rashmi et al., 2019).

et al. (2019) established an analytical model to evaluate the performance of semi-transparent PVT with inverted absorber (IA) FPC (SPVT-IAFPC) (Fig. 4) from exergetic point of view for three cases: conventional inverted absorber FPC, partially covered SPVT-IAFPC air collector, and fully covered SPVT-IAFPC-air collector. The authors simulated the conducted model using MATLAB software to demonstrate the effect of mass flow rate on the outlet temperature, rate of thermal efficiency, cell temperature and electrical efficiency, and to evaluate the thermal energy performance with the global exergy efficiency.

PTC is made up of a parabolic concentrator of reflective material, supporting structure, tracking system, metal glazed receiver tube adjusted at the focus of the concentrator and working fluid passing through it. Karathanassis et al. (2019) conducted a theoretical dynamic model to demonstrate the impact of several geometrical system components, materials and functional parameters on the energetic and exergetic performance of a parabolic trough PVT system. It was concluded that the exergetic efficiency of the system augmented from 12% to 24% when rising the optical efficiency of the parabolic trough and the electrical efficiency of the PVM to 75% and 25% respectively. Calise and Vanoli (2012) designed and established thermal modeling of a novel concentrating PVT system where the structure was integrated from an existing prototype in the literature. The system is made up of parabolic trough collector and triangular receiver containing fluid channel, two PV layers facing the concentrator and absorber at the top of the receiver. Fig. 5 illustrates the layout of the system. The model was simulated to investigate the electrical and thermal performance of the system. It was concluded that the thermal and electrical efficiencies mitigate with the alleviation in the incident beam radiation. Also, the authors developed a detailed finite volume model of the parabolic trough PVT to assess its thermodynamic and electrical parameters.

Furthermore, Valizadeh et al. (2019) conducted thermal modeling to evaluate the exergy performance of the system studied by Calise and Vanoli (2012). The developed thermal modeling is simulated by varying several design parameters such as channel diameter, fluid velocity, receiver width, collector length and boundary conditions such as fluid inlet temperature, ambient temperature and incident beam radiation. Yazdanifard et al. (2017) conducted thermal modeling on parabolic trough PVT system to study its energetic and exergetic performance in laminar and turbulent flows (Fig. 6). The thermal modeling was validated using experimental results existing in the literature and then it is used to investigate the effect of concentration ratio, pipe diameter, and length on the performance of the studied system. The

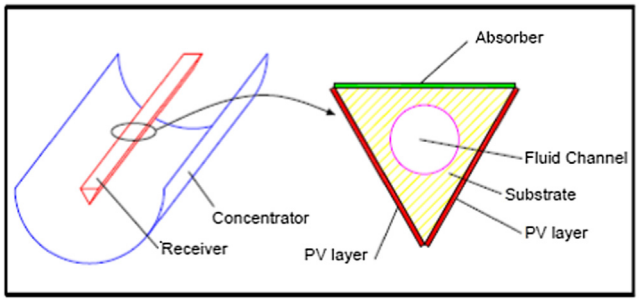


Fig. 5. Parabolic trough PVT system investigated by (Calise and Vanoli, 2012).

outcomes revealed that the total energy efficiency is mitigated by approximately 1.09% and 9.33% in the turbulent and laminar regime respectively when pipe length rises from 0.5 to 5 m, while the exergy efficiency augments by about 10.37% and 33.65% in the turbulent and laminar regime respectively. The authors also compared the system performance with that of a flat plate PVT system.

CPC is composed of two meeting parabolic concentrators, transparent cover from the top, absorber, and insulation around the fluid passages. There are two main types of compound parabolic concentrators which are: symmetric and asymmetric. Also, two types of absorbers are utilized: fin type with a pipe and tubular absorbers. Lu et al. (2018) designed and developed a Building Facade Integrated Asymmetric Compound Parabolic Photovoltaic concentrator (BFI-ACPPV) system. A comprehensive indoor test was performed to assess the electrical and thermal performance of the system as well as the parameters that influence the power output of the system. It was observed, from the experimental results, that the power output per unit solar cell area is raised by a factor of two by utilizing the BFI-ACPPV system in comparison to the conventional PV system. Jaaz et al. (2017) reviewed the various kinds of compound parabolic concentrator PVT system including the design benefits and barriers. The authors compared the utilized materials, the range of acceptance angle, optical tolerance and efficiency. Tiwari et al. (2018a) investigated and analyzed the operating parameters of a fully covered PVT-CPC based on applying the energy balance on each part of the system. The system is shown in Fig. 7 where it is mainly made up of solar cells, absorber plate, concentrator and flowing fluid. The system performance was compared to PVT-FPC, conventional flat plate-CPC and conventional FPC. It was concluded that the PVT compound parabolic concentrator and the PVT-FPC are appropriate to be used in an absorption refrigeration system.

Also, Tiwari et al. (2020) carried out an hourly thermodynamic investigation on solar-powered organic Rankine cycle integrated with a hybrid reversed absorber partially covered PVT-CPC system. The aim of this study was to enhance the performance of the system to compensate for the increasing energy demand and alleviate environmental problems. The collector output temperature, combined thermal and exergy efficiency and the heat gain in evaporator were evaluated for different concentration ratios. It was observed that at 13:00hr and for concentration ratio equal to six, the highest collector outlet temperature, heat gain, combined thermal efficiency, and exergy efficiency were 125.45 °C, 3.21 × 10^5 W, 7.8%, and 14.38% respectively.

PDC mainly comprises a parabolic dish, absorber, flexible pipe and tracking axis. The parabolic dish is made up of reflective mirrors that direct sunlight towards the receiver which a container of water. Sharaf and Orhan (2018) designed and analyzed the optical and thermodynamic performance of two configurations of PDC-PVT: in parallel and in series thermal configurations. The two arrangements are evaluated from the exergo-economic and environmental point of view and then compared. It was concluded that the thermally-parallel design supplies significant electrical and thermal outputs and it is favorable for applications that request simultaneously high electricity and temperature thermal energy. On the other hand, the thermally-series configuration is less flexible and is suitable for applications of low temperature thermal energy. Kribus et al. (2006) designed and assessed a new miniature PDC-PVT system presented in Fig. 8. The authors analyzed the manufacturing cost and the cost of energy required for domestic water heating applications in addition to the electrical and thermal performance. The results revealed that the miniature parabolic dish PVT system can be constructed at a low competitive cost and it has relatively high thermal and electrical efficiencies in comparison to the flat plate PVT system.

Buonomano et al. (2013) designed and analyzed the thermodynamic performance of a parabolic dish PVT system which is mainly composed of parabolic dish concentrator, PV layer and fluid channel, metallic substrate and thermal insulation (Fig. 9). The authors conducted and simulated a mathematical model which enables to compute the

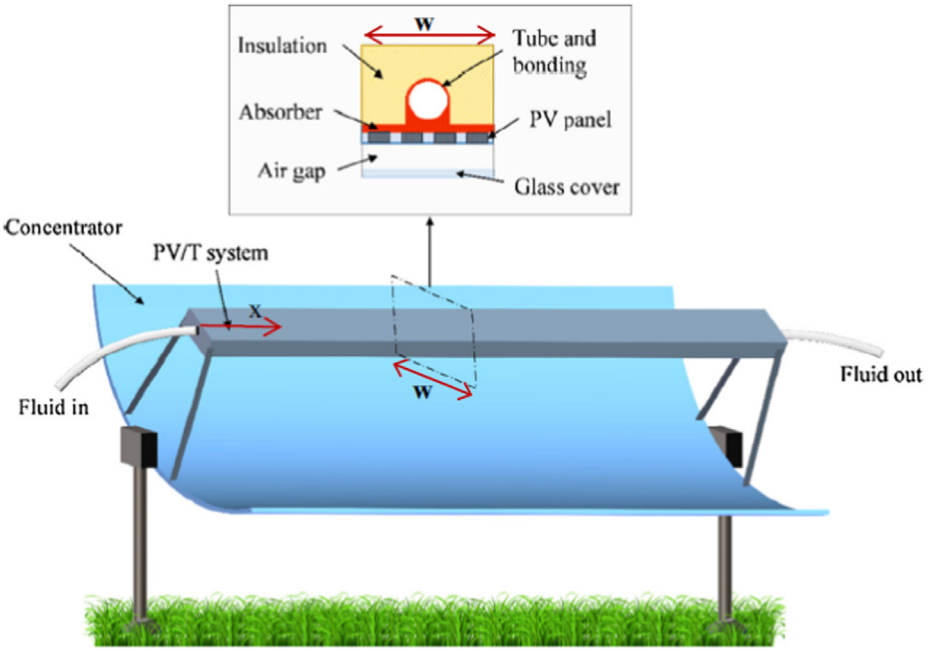


Fig. 6. Parabolic trough PVT system investigated by (Yazdanifard et al., 2017).

temperature of the system main components and the main energy flows such as useful thermal and electrical energy and the losses by convection and by radiation. Renno and Petito (2013) implemented a theoretical modeling of a parabolic dish PVT system that aims to size the system and assess its thermal and electrical operation in providing cooling, heating, and electricity to domestic application.

LFR is composed of an array of linear mirrors that track the sun, receiver tube that is fixed on the top of a tower and heat transfer fluid. Renno and Petito (2016) conducted experimental and theoretical analyses on the linear Fresnel PVT system to assess the electrical parameters as well as the thermal parameters such as the concentration factor, the fluid temperature and the cell temperature at distinct operating conditions. The aim of this analysis is to compute the potential energy generated by the studied system. Fig. 10 shows a schematic of the studied system.

Karimi et al. (2017) analyzed the performance of a pure thermal and PVT solar system using a concentrated linear Fresnel lens. The results showed that the total efficiency of the pure and PVT systems achieved 46.6% and 53% respectively. Wu et al. (2012) developed extensive indoor experimental testing to evaluate the heat loss from a Fresnel lens concentrator PVT collector where the concentration ratio is 100. The tests were simulated for solar radiation ranging from 200 to 1000 W/m^2, natural or forced convection and distinct ambient air temperature. It was shown that solar cell temperature rises when the solar radiation intensity increases. Sonneveld et al. (2011) developed and tested a linear Fresnel lens PVT collector with greenhouse for agriculture. The

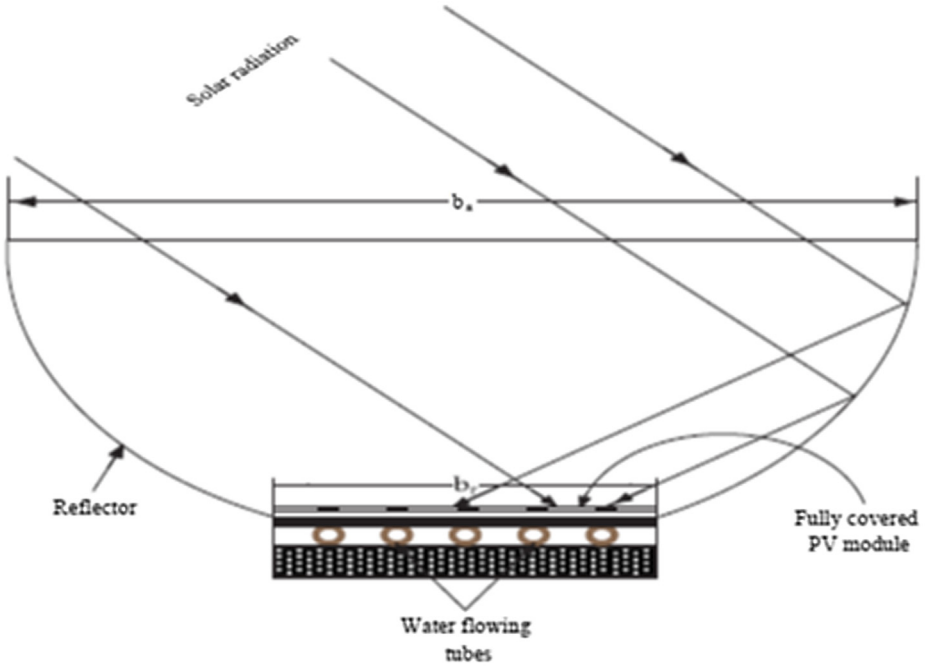


Fig. 7. Compound parabolic concentrator PVT system (Tiwari et al., 2018).

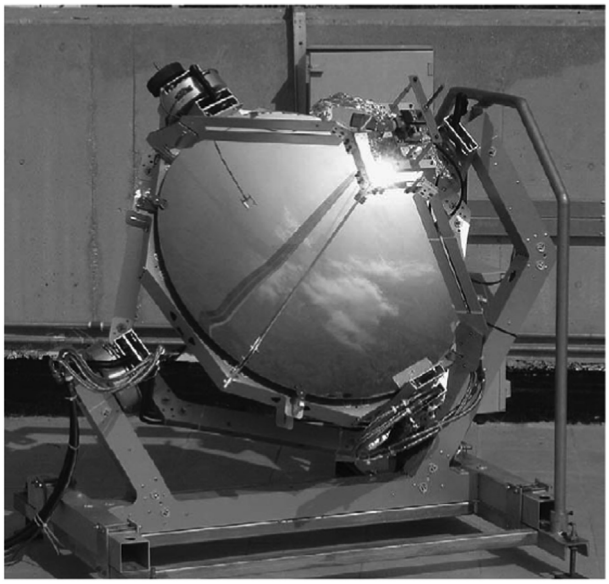


Fig. 8. Parabolic dish PVT collector designed by (Kribus et al., 2006).

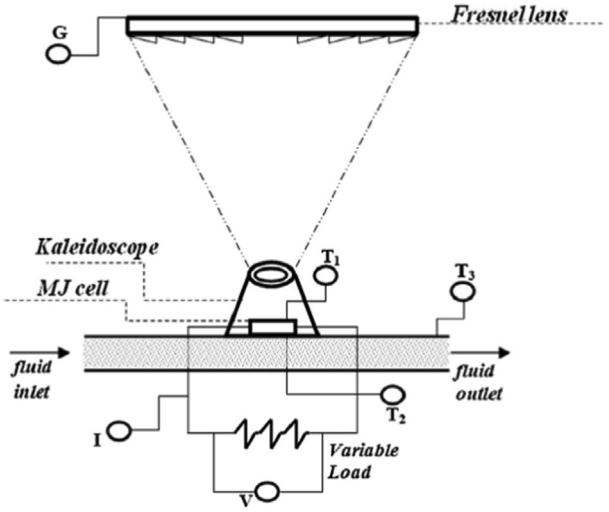


Fig. 10. Fresnel lens PVT system (Renno and Petito, 2016).

results revealed that the electricity generated was 2% greater than the energy consumed by the tracking system. It was obtained that the thermal and electrical efficiencies of the system are 56% and 11% respectively.

Table 2 presents recent studies that were performed on PVT hybrid systems according to the type of solar collector used.

Analysis of Table 2 yields to the following concluding points:

- All solar collectors including the concentrated and non-concentrated types can operate at high solar radiation (about 1000 W/m^2)
- The selection of the solar collector depends on the type of application where each one requires certain range of outlet temperature.
- Concentrated type of STC; mainly parabolic trough and linear Fresnel are the most commonly utilized types in PVT systems.
- The electrical efficiency is negligibly affected by the type of the STC.

3.3. Working fluid

Photovoltaic thermal systems can be classified relying on the working fluid circulation (natural or forced circulation), working fluid type (water, air or water/air), location of fluid channel (above or below PV layer) and fluid entry (single or multiple). Forced circulation of the working fluid is more efficient than natural flow circulation. However, the required power by the fan mitigates the net electric power generated by PVT system.

Air was firstly used as a working fluid in PVT systems with natural circulation where after that forced convection of air is adopted. PVT air collector has similar structure to solar air collector in which the thermal absorber sheet is replaced by PV layer. In such collector, air passes behind the PV layer and removes heat from it to be utilized in several applications such as drying or space heating. PVT air system has no leakage and freezing problems and it has low cost. However, air is not preferred as a heat transfer working fluid since it has low density, thermal conductivity and specific heat.

Barone et al. (2019a) established a dynamic simulation model to assess the energy performance and economic feasibility of low cost air based PVT collector. Experimental tests were carried out to validate the results of model simulation.

Fudholi et al. (2019) proffered a review to study the sustainability index and exergy of PVT air system. The authors established a model to investigate the performance of the system where the results are verified by conducting experimental tests.

Numan and Kaya (2019) demonstrated the energy and exergy analyses for PVT air collector. Using ANSYS Fluent, the authors investigated the distribution of temperature along the panel surface and the air velocity of the cooling channel. The results were compared to experimental results.

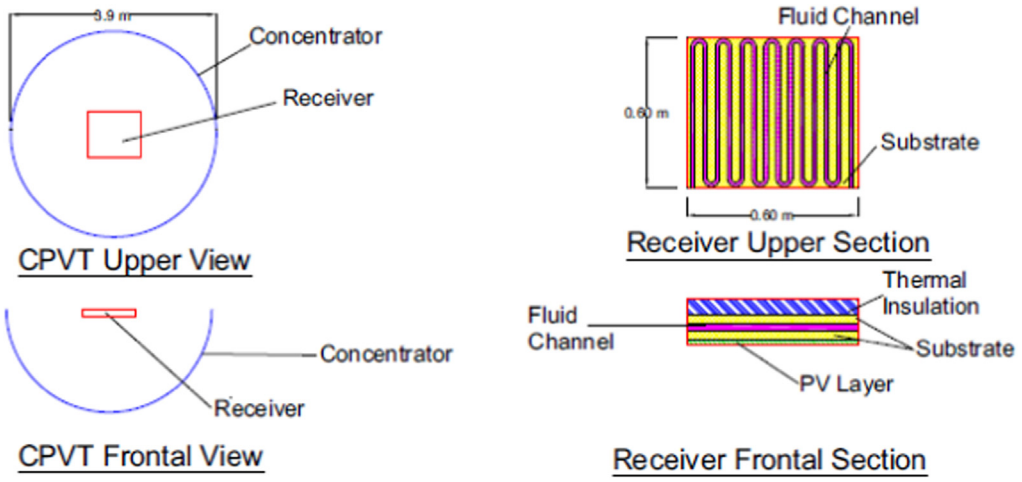


Fig. 9. Parabolic dish PVT collector investigated by (Buonomano et al., 2013).

Table 2
Studies performed on hybrid systems according to the type of solar collector.

St. Nb	Reference(Author/s, Year)	Thermal collector	Operating conditions	Efficiency			Type of study
				Thermal	Electrical	Total	
	Ibrahim et al. (2009a)	• Flat plate	• Solar radiation of 800 W/m^2 • Ambient temperature of 25 °C	45%	10%	55%	• Experimental
	Bernardo et al. (2011)	• Parabolic trough	• Incident beam radiation of 997 W/m^2 • Water inlet temperature of 28 °C	45%	6.4%	–	• Experimental
	Chemisana et al. (2011)	• Fresnel • Compound parabolic	• Solar irradiance between 7000 and 10000 W/m2	65%	–	–	• Experimental • Theoretical
	Sonneveld et al. (2011)	• Fresnel	• Solar radiation of 630 W/m2 and 792 W/m2	56%	11%	67%	• Experimental
	Dupeyrat et al. (2011)	• Flat plate	• Total radiation of 960 W/m2 • Ambient temperature varied between 30 and 33 °C • Wind velocity of 3 m/s	79%	8.8%	–	• Theoretical • Experimental
	Calise and Vanoli (2012)	• Parabolic trough	• Inlet water temperature of 70 °C • Ambient and sky temperature of 25 °C • Wind velocity of 5 m/s • Total and beam radiation of 1000 W/m2 and 800 W/m2 respectively	56%-63%	20%–23%	77%-83%	• Theoretical
	Buonomano et al. (2013)	• Parabolic dish	• Inlet fluid temperature of 150 °C • Ambient and sky temperature of 25 °C • Wind velocity of 5 m/s • Total and beam radiation of 1000 W/m2 and 800 W/m2 respectively	27%-58%	18%-24%	45%-82%	• Theoretical
	Renno and Petito (2013)	• Parabolic dish	–	67%-70%	19%-21%	86%-91%	• Theoretical
	Del Col et al. (2014)	• Parabolic trough	• Inlet water temperature ranges between 20 and 80 °C • Beam radiation of 1000 W/m2	41.6%	23.1%	64.7%	• Theoretical • Experimental
	Hussain and Lee (2015)	• Fresnel	• Solar radiation of 900 W/m2 • Fluid inlet temperature of 25 °C	68%	–	–	• Theoretical • Experimental
	Xu et al. (2015)	• Fresnel	• Ambient temperature between 15 and 17 °C • Beam radiation of 700 W/m2	54%	28%	84%	• Experimental
	Yew et al. (2015)	• Compound parabolic • Parabolic dish	• Solar irradiance of 1000 W/m2 • Default temperature of 25 °C	34.1%-38.7%	20%-25%	–	• Theoretical
	Mi et al. (2016)	• Fresnel	• Solar irradiance of 900 W/m2 • Ambient temperature of 25 °C	–	–	–	• Experimental • Theoretical
	Feng et al. (2016)	• Fresnel	• Solar irradiation between 925 and 1006 W/m2	45%	18%	55%	• Experimental
	Cui et al. (2016)	• Fresnel	• Ambient temperature of 27 °C • Water inlet temperature of 20 °C • PV cell temperature between 21 and 35 °C • Total solar irradiance of 1000 W/m2	–	–	25%-27%	• Theoretical
	Mojiri et al. (2016)	• Fresnel	–	50%	3.8%	–	• Theoretical • Experimental
	Widyolar et al. (2017)	• Parabolic trough • Compound parabolic	• Beam radiation of 700 W/m2 • Fluid inlet temperature between 100 and 500 °C	37%	8%	–	• Theoretical • Experimental
	Ceylan et al. (2016)	• Compound parabolic	• Ambient temperature between 20 and 40 °C • Total radiation of 1000 W/m2	–	11%	–	• Theoretical • Experimental
	Gürel (2016)	• Compound parabolic	• Total radiation of 1000 W/m2 • Cell temperature of 25 °C	–	–	11%-16%	• Experimental
	Xu et al. (2016)	• Fresnel	• Ambient temperature of 25 °C • Wind velocity of 4 m/s • Beam radiation of 1000 W/m2	60%	28%	–	• Theoretical
	Atheaya et al. (2016b)	• Compound parabolic	• Beam radiation between 100 and 600 W/m2 • Total radiation between 150 and 750 W/m2	–	6.5%-13.4%	–	• Theoretical
	Karathanassis et al. (2017)	• Parabolic trough	• Ambient temperature between 15 and 35 °C • Wind velocity between 0 and 2 m/s • Beam radiation between 760 and 970 W/m2	44%	6%	50%	• Experimental
	Karimi et al. (2017)	• Fresnel	• Solar radiation of 920 W/m2 • PV cell temperature of 20 °C	43.6%	14.5%	53%	• Experimental
	Burhan et al. (2017)	• Flat plate • Fresnel • Parabolic dish	–	–	22%	23%	• Theoretical
	Srivastava and Reddy (2017)	• Parabolic trough • Compound parabolic	• Fluid inlet temperature and ambient temperature of 27 °C • Beam radiation of 650 W/m2	69.68%	18.91%	85.82%	• Theoretical • Experimental

(continued on next page)

Table 2 (*continued*)

St. Nb	Reference(Author/s, Year)	Thermal collector	Operating conditions	Efficiency			Type of study
				Thermal	Electrical	Total	
	Tripathi and Tiwari (2017)	• Compound parabolic	• Ambient temperature between 25 and 34 °C • Beam radiation between 300 and 860 W/m2	37.1%-37.5%	2.7%-3.9%	–	• Experimental • Theoretical
	Yazdanifard et al. (2017)	• Parabolic trough • Flat plate	• Fluid inlet temperature and ambient temperature of 25 °C • Solar radiation of 700 W/m2 • Wind velocity of 1.5 m/s	52.58%-56.37%	7.12%-7.94%	61%-63.5%	• Theoretical
	Proell et al. (2017)	• Compound parabolic • Flat plate	• Ambient temperature between 17 and 29 °C • PV cell temperature of 22.9 °C • Total radiation of 1000 W/m2	34%	9%	–	• Experimental
	Chong et al. (2017b)	• Parabolic dish	• Direct normal irradiance of 1000 W/m2	–	17%	–	• Experimental
	Chong et al. (2017a)	• Parabolic dish	• Solar irradiance of 1000 W/m2	–	–	16.1%-17.4%	• Experimental • Theoretical
	Schmitz et al. (2017)	• Parabolic dish	• Fluid inlet temperature of 85 °C	–	26.6%	–	• Theoretical
	Lu et al. (2018)	• Compound parabolic	• Incident radiation intensity of 280 W/m2 • Ambient temperature of 23.9 °C	–	14.8%	–	• Theoretical • Experimental
	Sakellariou and Axaopoulos (2018)	• Flat plate	• Wind speed between 0 and 6 m/s • Ambient temperature between 24 and 28 °C • Incident solar irradiance between 200 and 900 W/m2	–	–	–	• Theoretical • Experimental
	Tiwari et al. (2018a)	• Compound parabolic • Flat plate	• PV cell temperature of 25 °C • Total solar radiation of 1000 W/m2	–	• 0.1% • 0.11%	–	• Theoretical
	Aoul et al. (2018)	• Flat plate • Fresnel	• Ambient temperature between 21 and 45 °C • Wind speed between 0 and 5 m/s	24%-36%	20%-24%	30%-47%	• Experimental
	Jaaz et al. (2018)	• Compound parabolic	• Ambient temperature of 33 °C • Solar radiation of 800 W/m2	81%	13.75%	91%	• Experimental
	Soltani et al. (2018)	• Parabolic trough	• Ambient temperature of 36.2 °C • Water inlet temperature of 28 °C • Solar radiation of 900 W/m2	57%	–	–	• Theoretical
	Adam et al. (2019)	• Fresnel • Parabolic trough	• Ambient temperature of 25 °C • fluid inlet temperature of 20 °C • Direct normal irradiance of 1000 W/m2	55%-70%	5%-18%	66%-85%	• Theoretical
	Bellos and Tzivanidis (2019)	• Parabolic trough	• Solar irradiation of 1000 W/m2 • Reference temperature of 25 °C	46.84%	6.6%	–	• Theoretical
	El-samie et al. (2019)	• Compound parabolic	• Reference temperature of 25.15 °C	48%-50.6%	6.9%-7.3%	80%-89%	• Theoretical
	Shittu et al. (2019)	• Flat plate	• Ambient temperature between 15 and 55 °C • Wind speed between 0 and 7 m/s	–	–	16%-17%	• Theoretical
	Yu et al. (2019)	• Flat plate	• Ambient temperature between 21.6 and 23.6 °C • Wind speed between 2.2 and 3 m/s • Solar irradiance between 880 and 940 W/m2 • PV cell temperature between 20 and 45 °C	24.2%-25.2%	11.8%	36%-37%	• Theoretical • Experimental
	Zhang et al. (2019)	• Compound parabolic	• Ambient temperature of 18 °C • Wind speed of 0.1 m/s • Solar irradiance of 900 W/m2 • Inlet fluid temperature of 14.5 °C	56.7%	11.8%	66.4%	• Theoretical
	Valizadeh et al. (2019)	• Parabolic trough	• Ambient temperature of 25 °C • Sky temperature of 19 °C • Wind speed of 5 m/s • Beam radiation of 800 W/m2 • Total radiation of 1000 W/m2 • Inlet fluid temperature of 70 °C	59%-67%	15%–22%	–	• Theoretical
	Bamroongkhan et al. (2019)	• Parabolic dish	• Solar intensity between 300 and 826 W/m2 • Ambient temperature around 30 °C	–	19.65%	–	• Theoretical
	Wang et al. (2019)	• Fresnel	• Ambient temperature between 30 and 33 °C • Wind speed of 3 m/s • Solar radiation of 960 W/m2	79%	8.7%	88%	• Theoretical

Fan et al. (2018) presented a novel design for PVT solar air heating which aims to optimize the thermal energy utilized and the gained electrical energy. The authors studied several parameters that are related to geometry, construction materials and operation. Uncertainty study was also conducted which revealed that the useful thermal efficiency ranges between 48.8% and 56.9%.

Aste et al. (2008) overviewed the development and research program which was developed at the Politecnico di Milano by presenting its experimental and theoretical results. The program was carried out on the development, design and monitoring the performance of air

Table 3
Theoretical and experimental works on water or/and air PVT systems.

St. Nb	Reference(Author/s, Year)	Working fluid	Operating conditions	Efficiency Thermal	Electrical	Total	Type of study
	Hung et al. (2001)	Water	• Total radiation between 200 and 900 W/m2 • Ambient temperature between 25 and 35 °C	44.5%	9.1%	53.5%	• Experimental
	Zondag et al. (2003)	Water	• Ambient temperature of 20 °C • Sky temperature of 4 °C • Incident angle 45° • Wind velocity of 1 m/s • Total radiation of 800 W/m2	52%-58%	8.1%-9.7%	–	• Theoretical • Experimental
	Solanki et al. (2009)	Air	• Solar radiation of 600 W/m2 • Inlet fluid temperature of 38 °C	42%	8.4%	51%	• Theoretical • Experimental
	Aste et al. (2017)	Water	–	22%-29%	6%-14%	–	• Theoretical
	Ibrahim et al. (2009b)	Water	• Solar radiation of 800 W/m2 • Wind velocity of 1 m/s • Ambient temperature of 26 °C	50.12%	11.98%	68%	• Theoretical
	Xu et al. (2011)	Water	• Solar radiation between 500 and 900 W/m2 • Wind velocity of 0.75 m/s • Ambient temperature between 34.2 and 38.5 °C	–	17.5%	–	• Experimental
	Bambrook and Sproul (2012)	Air	• Solar radiation of 800 W/m2 • Ambient temperature of 17 °C	28%-55%	10.6%-12.2%	–	• Experimental
	Kim et al. (2014)	Water	–	30%	17%	47%	• Experimental
	Alobaid et al. (2017)	Water	–	–	10–35%	–	• Theoretical
	Fan et al. (2018)	Air	• Wind velocity between 0 and 8 m/s • Sky temperature between 10 and 22 °C • Beam radiation up to 810 W/m2	48.8%-56.9%	–	–	• Theoretical • Experimental
	Kazemian et al. (2018)	Water	• Total radiation between 400 and 580 W/m2 • Ambient temperature between 28 and 32 °C	63.37%-70.89%	13.15%-14.35%	76%-84%	• Theoretical • Experimental
	Tiwari et al. (2018b)	Air	• Ambient temperature between 35 and 43 °C • Total radiation between 100 and 900 W/m2 • Incident angle of 30°	11.26%	26.68%	56.3%	• Theoretical
	Maatallah et al. (2019)	Water	• Ambient temperature between 32 and 38 °C • Total radiation between 390 and 560 W/m2	26.87%	13.72%	40.59%	• Experimental
	Misha et al. (2019)	Water	• Inlet water temperature of 26 °C • Solar irradiance of 600, 800 and 1000 W/m2 • Wind velocity of 1 m/s • Ambient temperature of 30 °C	59.6%	11.71%	–	• Theoretical • Experimental
	Senthilraja et al. (2019)	WaterAir	• Ambient temperature between 34 and 38.6 °C • Wind velocity between 0.6 and 2.3 m/s • Total radiation between 420 and 940 W/m2	33.8%31.1%	8.5%8%	42%38%	• Theoretical • Experimental

based PVT collector. Also, the research included a simulation model for predicting and evaluating the performance of air PVT system.

Bambrook and Sproul (2012) studied the parameters that affect thermal performance of PVT air collector to determine the optimum design that suits the climate. The authors designed and conducted experiments on PVT system. The experimental results clarified that at midday the thermal and electrical efficiencies are in the range of 28–55% and 10.6–12.2% respectively.

Solanki et al. (2009) designed, fabricated and evaluated the performance of air based PVT collectors by performing indoor testing procedure. The authors established a steady state thermal model by applying energy balance equations. The results showed that the maximum thermal and electrical efficiencies achieved by PVT air collector are 42% and 8.4% respectively.

On the other side, water is more efficient than air since it has higher thermal properties than air, but the cost of leakage detector and maintenance, the electrically insulated thermal device and the water circulation pipes make it more expensive compared to PVT air system. Such system is used for space heating, desalination and water heating.

Hung et al. (2001) assessed the performance of solar water based PVT collector and compared it to traditional solar water heater. It was observed that the maximum thermal and electrical efficiencies of PVT water collector are 44.5% and 9.1% respectively.

Barone et al. (2019b) developed a dynamic simulation model to assess the energy, economic, and environmental performance of a low cost water based PVT collector. The authors designed, build and carried out experimental tests on a prototype of air water PVT collector to validate the results of the model.

Maatallah et al. (2019) compared the performance of water based PVT panel using phase change material with conventional PV panel by performing experimental tests under the weather conditions of Kottayam, India. It was observed that the electrical efficiency of PVT panel augmented by 17.33% in comparison to the traditional PV technology. The thermal and electrical efficiencies of PVT system are 26.87% and 13.72% respectively.

Yazdanpanahi et al. (2015) accomplished numerical and experimental demonstration on the exergy efficiency of PVT water collector. The authors performed experimental testing where the measured parameters comprise the wind speed, intensity of solar radiation, open circuit voltage and solar short circuit current, maximum power point voltage and maximum power point current, ambient temperature, fluid inlet and outlet temperature. Also, they conducted numerical simulation using four parameter current voltage model and thermal model of one dimensional steady state. The results revealed similarity between the numerical and experimental results.

Shyam et al. (2015) analyzed and compared the performance of two water collectors partially covered with PVM where the two collectors differs by the position of the PVM whether it is at the lower or upper portion of the collector. It was shown from the analytical analysis that at moderate mass flow rate and for large number of PVT water based collectors connected in series, both configurations have close results.

Aste et al. (2015) designed a covered PVT water collector which is composed of thin film PV cell integrated with flat plate absorber of roll-bond configuration. A simulation model was developed based on various mathematical equations. The aim of the model is to assess the performance of the designed PVT water based collector.

Misha et al. (2019) designed and implemented a PVT water system. The authors conducted a numerical simulations using ANSYS 19.2 to

Table 4
Theoretical and experimental works on PVT systems according to glazing.

St. Nb	Reference(Author/s, Year)	Glazing	Operating conditions	Efficiency Thermal	Electrical	Total	Type of study
	Hung et al. (2001)	No glazing	• Total radiation between 200 and 900 W/m2 • Ambient temperature between 25 and 35 °C	44.5%	9.1%	53.5%	• Experimental
	Zondag et al. (2003)	No glazingSingle glazingDouble glazing	• Ambient temperature of 20 °C • Sky temperature of 4 °C • Incident angle 45° • Wind velocity of 1 m/s • Total radiation of 800 W/m2	52%58%58%	9.7%8.9%8.1%	–	• Theoretical • Experimental
	Solanki et al. (2009)	Single glazing	• Solar radiation of 600 W/m2 • Inlet fluid temperature of 38 °C	42%	8.4%	51%	• Theoretical • Experimental
	Aste et al. (2017)	Single glazingNo glazing	–	29%22%	6%14%	–	• Theoretical
	Ibrahim et al. (2009b)	Single glazing	• Solar radiation of 800 W/m2 • Wind velocity of 1 m/s • Ambient temperature of 26 °C	50.12%	11.98%	68%	• Theoretical
	Xu et al. (2011)	Single glazing	• Solar radiation between 500 and 900 W/m2 • Wind velocity of 0.75 m/s • Ambient temperature between 34.2 and 38.5 °C	–	17.5%	–	• Experimental
	Bambrook and Sproul (2012)	No glazing	• Solar radiation of 800 W/m2 • Ambient temperature of 17 °C	28%-55%	10.6%-12.2%	–	• Experimental
	Kim et al. (2014)	No glazing	–	30%	17%	47%	• Experimental
	Slimani et al. (2016)	No glazingSingle glazingDouble glazing	• Solar radiation between 190 and 700 W/m2 • Inlet air temperature between 305 and 318 °C	21.19%40.77%44.41%	10.73%10.33%10.56%	51.02%69.47%74%	• Theoretical • Experimental
	Fan et al. (2018)	Single glazing	• Wind velocity between 0 and 8 m/s • Sky temperature between 10 and 22 °C • Beam radiation up to 810 W/m2	48.8%-56.9%	–	–	• Theoretical • Experimental
	Kazemian et al. (2018)	Single glazingNo glazing	• Total radiation between 400 and 580 W/m2 • Ambient temperature between 28 and 32 °C	70.89%63.37%	13.15%14.35%	84%76%	• Theoretical • Experimental
	Maatallah et al. (2019)	No glazing	• Ambient temperature between 32 and 38 °C • Total radiation between 390 and 560 W/m2	26.87%	13.72%	40.59%	• Experimental
	Misha et al. (2019)	No glazing	• Inlet water temperature of 26 °C • Solar irradiance of 600, 800 and 1000 W/m2 • Wind velocity of 1 m/s • Ambient temperature of 30 °C	59.6%	11.71%	–	• Theoretical • Experimental

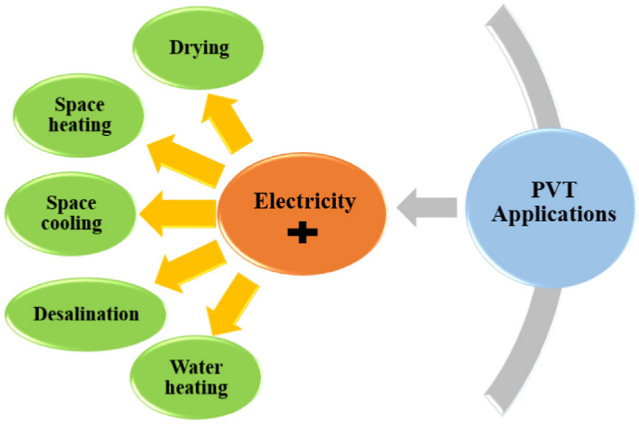


Fig. 11. Main applications used in PVT hybrid system.

evaluate the outlet temperature of water and the surface temperature of PV layer for various values of solar irradiation (600, 800 and 1000 W/m^2), different mass flow rate (2, 4 and 5 L/min) and 26 °C water inlet temperature. The simulation results were validated by experimental tests. It was revealed that the maximum electrical efficiency achieved by PVT system is 11.71% at 6 L/min mass flow rate. Also, it was concluded that the thermal efficiency augments with the increase in the solar irradiation, mass flow rate and alleviation in the variance between water outlet and inlet temperature.

Senthilraja et al. (2019) assessed and compared the performance of air and water based PVT collector for hydrogen production application. Experimental testing was performed in which a PVM was considered as a reference in comparing the two systems. It was obtained that at noon and for 0.011 kg/s mass flow rate, the electrical and thermal efficiencies are 8% and 31.1% for air based PVT collector and 8.5% and 33.8% for water based PVT collector respectively.

Table 3 presents the theoretical and experimental studies on water or/and air PVT systems.

Analysis of Table 3 yields to the following concluding points:

- Water is utilized more than air in PVT systems
- Water based system is more restricted by the operating conditions than air based system due to the freezing of water at low operating temperatures.
- Higher electrical, thermal and total efficiencies are obtained in water based PVT system.

3.4. Glazing

Glazing is a vital component of PVT collector in which it protects the panels from various environmental impacts. On the other hand, glazing may prohibit absorption of part of solar radiations by the PV cells depending on its transmissivity. If the cooling fluid is at the top side of the PV layer, a glazing may be utilized which acts as a channel for the fluid. Glazing is essential for maximizing the produced thermal energy, but it is not favorable in case high amount of electrical energy is aimed to be generated.

Kazemian et al. (2018) proffered a comparison in the performance of unglazed and glazed water based PVT system. The obtained results declared that the thermal efficiency of glazed and unglazed system is 70.89% and 63.37% respectively while the electrical efficiency is 13.15% for glazed system and 14.35% for unglazed system. It was concluded that if the aim is to obtain higher overall efficiency, the glazed system is preferred, but if the aim is to produce more electrical energy the unglazed system is more suitable to be used.

Also, Guarracino et al. (2019) performed experimental study to compare the impact of glazing on the PVT collector performance. It was proved that in the glazed system, electrical efficiency is relatively 20% lower than that of the unglazed system. Also, it was shown that adding the glass cover yields to augment the thermal efficiency.

Besides, ***Zondag et al. (2003) assessed the performance of several flat plate PVT collectors. The results showed that the thermal efficiency is 52% for unglazed collector while 58% for single and double glazed collector, while the electrical efficiencies are 9.7%, 8.9% and 8.1% for unglazed, single and double glazed collector respectively.

Chaysaz et al. (2019) evaluated the performance of small-scale greenhouse PVT system with three different covers: glass, plastic and polycarbonate. It was detected that the highest overall system efficiency was achieved when utilizing glass cover and the lowest one was attained when using polycarbonate cover. This is because the glass, plastic and polycarbonate glazing have transmissivity of 90, 85 and 75% respectively.

Chow et al. (2009) conducted a study to specify if glass cover is appropriate to be used in thermosiphon water heating PVT system. The study was performed by relying on experimental results and legitimized numerical models. It was concluded that the glazing can be utilized if the aim is to maximize the thermal energy or the global energy produced.

Yazdanifard et al. (2016) studied the effect of collector length, pipes diameter and number, Reynolds number, solar irradiation and packing factor on the performance of glazed and unglazed flat plate water PVT collector in turbulent and laminar flow. The results showed that the energy efficiency is higher in collector with glazing while the exergy efficiency depends on the collector length, packing factor and Reynolds

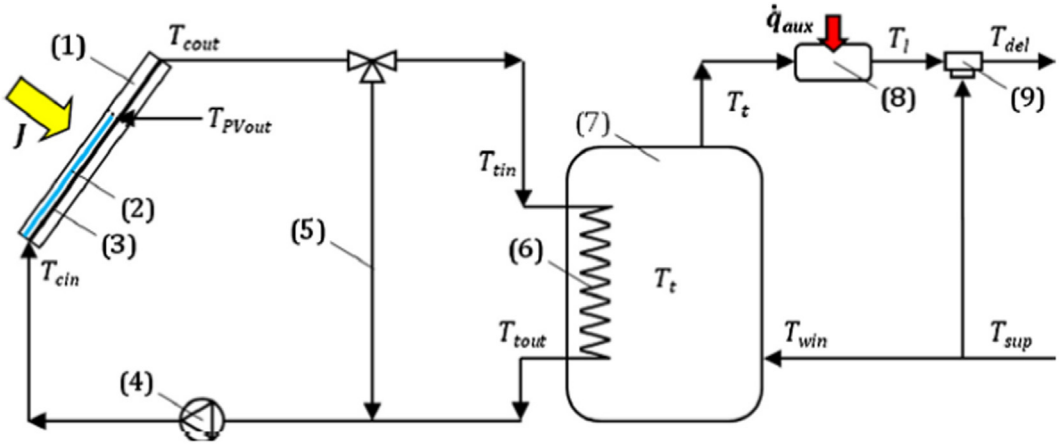


Fig. 12. PVT domestic hot water system (Herrando et al., 2014).

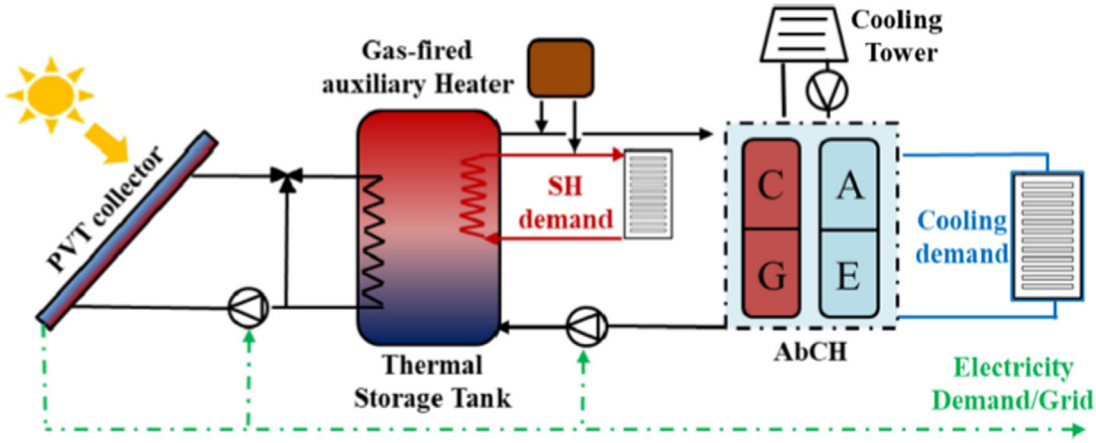


Fig. 13. Space cooling/heating PVT system (Herrando et al., 2019a).

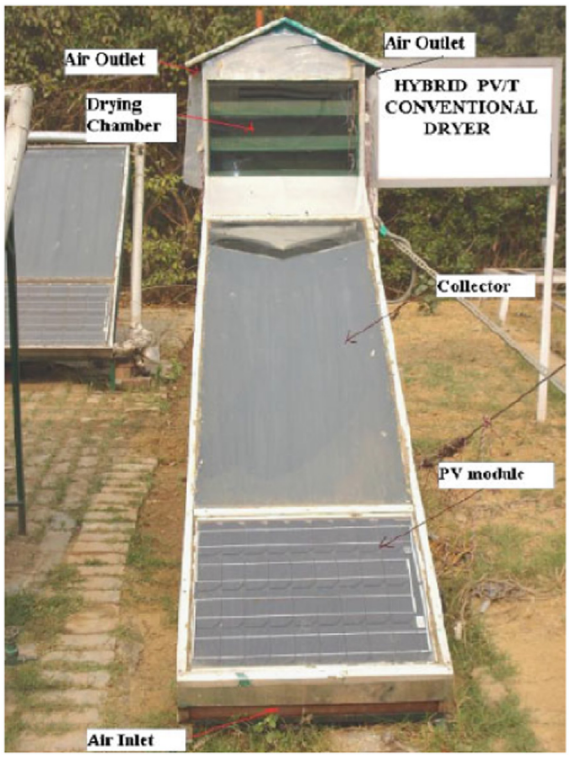


Fig. 14. Mixed solar dryer with PV module designed by (Tiwari et al., 2009).

number.

Aste et al. (2017) evaluated the effect of glazing on the performance of flat plate PVT collector by testing and comparing collectors with and without cover. It was shown that the electrical efficiency is 6% for covered collector and 14% with uncovered collector, while the thermal efficiency was 29% and 22% for covered and uncovered collector respectively.

Table 4 summarizes the theoretical and experimental works that were conducted during the last several years on solar PVT systems using single or double glazing or with no glazing.

Analysis of Table 4 yields to the following concluding points:

- The usage of glazing leads to increase the thermal efficiency and slightly decrease the electrical efficiency.
- Higher total efficiency of PVT system is obtained when utilizing glazing.
- Unglazed PVT system is more favorable if high electrical output is required.
- Glazing is more preferred to be used in PVT systems in case high thermal or overall system efficiency is required.
- Single glazing is more preferable than double glazing.

3.5. Thermal absorber

PVT system is generally constructed by attaching PV cells to the thermal absorber of a solar collector. Hence, thermal absorbers are the complement of solar cells in PVT system in which they absorb SE to exploit it. The thermal absorber has a vital impact on the thermal and electrical efficiency of PVT system. Thermal absorber is mainly located between the PVM and the fluid, so it has an important role in transferring the heat from PV layer to the fluid. The model of the thermal absorber depends on the type of solar collector used in PVT system. The main materials utilized in thermal absorbers are aluminum and copper.

Wu et al. (2016) proffered a review including the thermal absorbers of solar PVT system and their ways of integration with PVMs. The review describes the main types of PVT technology regarding their structure, efficiencies and applications. Also, the review comprises the types of thermal absorbers, their methods of integration, their advantages and disadvantages as well as the applications for several PVT modules.

Herrando et al. (2019b) investigated the potential of progressing hybrid PVT collectors with optimum performance, cost, weight and manufacturing simplicity. The authors proposed 26 alternative absorber-exchanger designs which are made up of various materials and assessed their thermal and electrical performance.

Singh et al. (2019) conducted a mathematical modeling to investigate the impact of mass flow rate and absorber shape factor in order to optimize the flat plate PVT system efficiency. It was concluded that for low and high mass flow rates, the curve grooves in the absorber plate are suitable to be used.

Ibrahim et al. (2009b) designed, investigated and compared seven novel design configurations of thermal absorber to study the influence of thermal absorber structure on the electrical and thermal performance of PVT system. It was revealed that the spiral flow configuration was the best choice for flat plate collector thermal absorber, where it achieved 50.12% thermal efficiency and 11.98% electrical efficiency.

Charalambous et al. (2011) conducted a new mathematical investigation to precise the best design of absorber plate which has least cost but higher efficiency. A steady state model was established to assess the optimum layouts performance. The results of the model were also validated by experimental test. It was obtained that the optimum configurations have comparable thermal and electrical efficiencies with the non-optimum designs which have greater mass and cost.

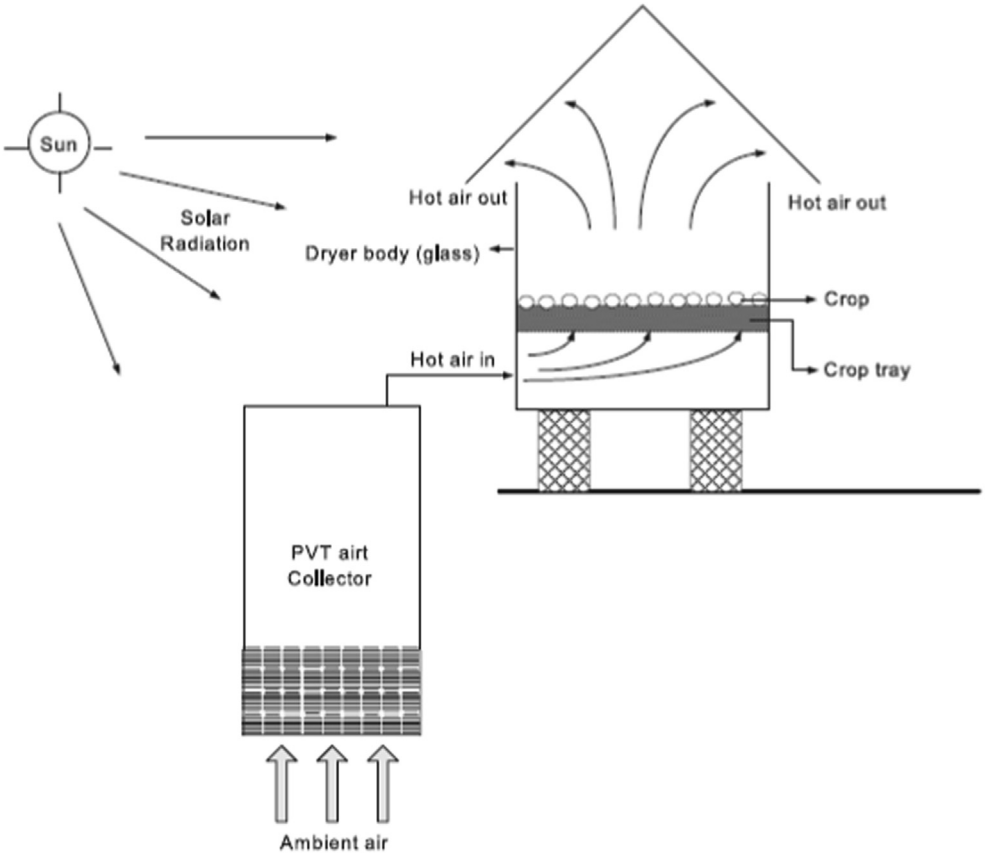


Fig. 15. Schematic of greenhouse dryer with air PVT collector designed by (Tiwari et al., 2018).



Fig. 16. Mixed solar dryer with PV module designed by (Fterich et al., 2018).

4. Applications

Mainly, PVT is used for the aim of producing electricity and heat. Heat energy can be exploited in various applications such as heating water (Li and Sun, 2018a, 2018b; Taboada et al., 2017; Zhang et al., 2014, 2013), space heating and cooling (Buonomano et al., 2018, 2017; Cremers et al., 2015; He et al., 2014; Herrando et al., 2018; Kazanci et al., 2014; Moaleman et al., 2018; Xu and Kleinstreuer, 2014), drying (Tiwari and Tiwari, 2017a, 2017b, 2016a, 2016b) and desalination (Ammous and Chaabene, 2015; Bilton, 2018; Calise et al., 2019; Kelley and Dubowsky, 2013; Lee et al., 2012; Li et al., 2019; Manokar et al., 2017; Modi and Shukla, 2018). Fig. 11 illustrates the main applications used in PVT hybrid system.

SWH is an application of solar energy which exploits thermal energy from solar radiation to heat water for domestic and industrial aims. Generally, SWH system is made up of three major components: STC, storage tank and heat transfer fluid. SWH systems are commonly classified into active and passive systems. Xu et al. (2011) established a new heat pump technology integrated with a parabolic concentrator PVT collector and carried out several experiments on it in Nanjing, China. It was detected that the novel system attained 4.8 mean Coefficient of Performance while water was heated from 30 °C to 70 °C during a sunny summer day with 17.5% electrical efficiency. Fu et al. (2019) conducted a comparative study of the performance of PVT-SWH systems with the direct-coupled photovoltaic pump, the conventional pump, and natural circulation by analyzing their performances through experimental tests. The results showed that the best thermal performance between the three systems is achieved by the PVT system with the photovoltaic pump. The natural circulation system attained net energy close to that attained by the photovoltaic pump, while the conventional pump system produced energy much less than the two systems. In addition, it was shown that the highest energy and exergy efficiencies are 69.63% and 12.84% respectively, obtained by the natural circulation system. Martin-Escudero et al. (2019) established a study to investigate the performance of a novel system that comprises heat pump coupled with a photovoltaic ventilated facade. The system is utilized for supplying heating and hot water. The authors established a mathematical model to evaluate the system outputs. They concluded that the PV panel provides about 70% of the electricity exploited by the heat pump and fans of the PV ventilated facade. Kim et al. (2014) implemented a study which analyzes the experimental performance of building PVT heating system. It was concluded, from the experimental results, that the average thermal and electrical efficiencies are 30% and 17% respectively. Herrando et al. (2014) established a model (Fig. 12) to evaluate

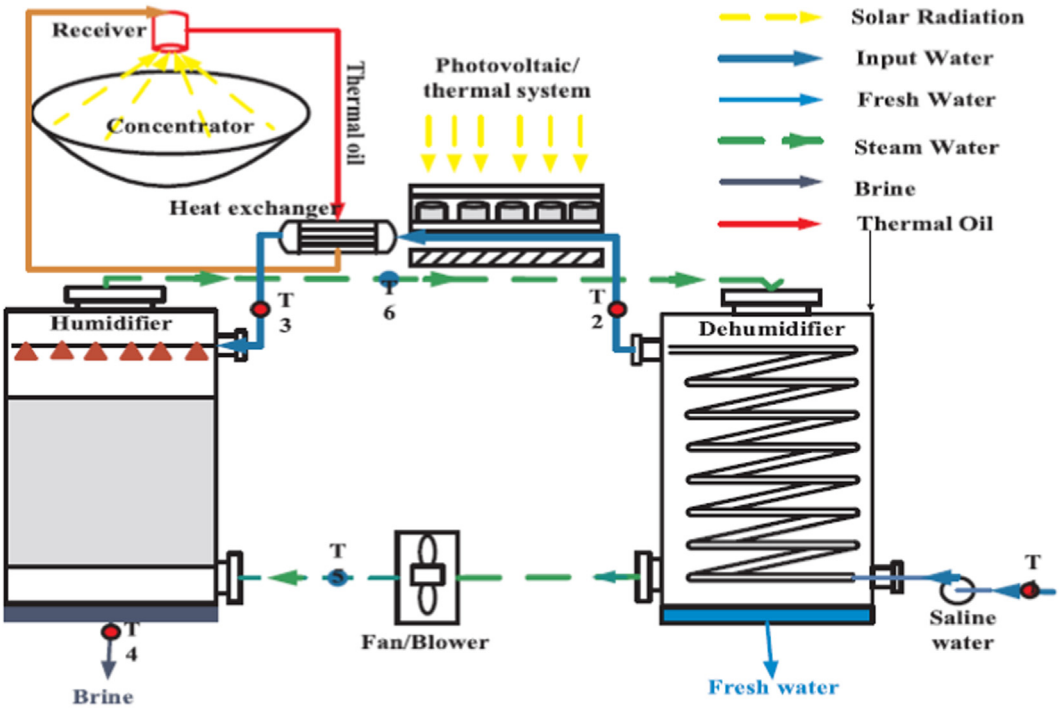


Fig. 17. Hybrid PVT desalination system (Rafiei et al., 2019).

the sustainability and performance of a PVT water heating system in a terraced house in London. The model is simulated by varying several design parameters of the studied collector, mainly the covering factor and the flow rate of the PVT collector, which enables to demonstrate the system overall performance. It was observed that for 20 L/h flow rate and completely covered collector, the hybrid PVT system can cover 36% and 51% of the annual global hot water and electricity demands respectively. Also, the hybrid PVT system serves in mitigating the CO_2 emission over 20 years lifetime by 16 tons which is 36% greater than the value saved by a PV system.

Space cooling/heating refers to the reduction/increase of the temperature of a room or workspace by removing/supplying heat from/to it. Solar cooling/heating system mainly includes a solar collector, heat rejection loop and thermal energy storage. Jani et al. (2018) presented a review on a solar solid desiccant vapor compression hybrid cooling system. The author described the principle of the solar solid desiccant hybrid cooling system and the recent progress in air-conditioning and built environment using hybrid cooling. Alobaid et al. (2017) presented a review on solar PVT absorption cooling systems to evaluate the latest progress in the solar collectors types used in solar absorption cooling systems. The review comprises experimental and computational previous works that studied the types of solar collectors, their efficiencies and performance. It was concluded from the comparison between vapor compression air conditioning and solar absorption cooling system that the usage of the latter saves 50% of the primary energy where the maximum electrical efficiency achieved by PVT is 35%. Noro and Lazzarin (2014) considered several distinct cooling systems and conducted analysis to determine the most favorable system. The authors performed dynamic simulation for two different climates during summer to assess the systems from energetic and economic point of view. Herrando et al. (2019a) developed a modeling methodology on hybrid PVT solar system for space heating and cooling and electricity generation (Fig. 13). The aim of this methodology is to assess the techno-economic performance of the system. It was observed that 16.3%, 20.9% and 55.1%, of the electricity, space heating and space cooling requirements, respectively, of the University Campus of Bari, can be provided by 1.68 MW system. The system payback period is 16.7 years which is 2.7 times greater than that of traditional PV system, and it can reduce 911 tons of CO_2 emission per year.

Wang et al. (2019) suggested and investigated the performance of a new PVT-CPC system for cooling, heating and power generation. The authors conducted thermodynamic models, compared and validated them from results of existing studies and then studied the impact of several factors on the system performance. The results showed that at design conditions, the energy and exergy efficiencies are respectively 61.8% and 27.1% in winter and 63.3% and 21.8% in summer. Also, Chen et al. (2019) studied a novel hybrid system for cooling, heating and power production using PVT-CPC collector to optimize and minimize its thermo-ecological cost. To achieve this aim, the authors proposed a methodology to optimize the configurations and installations of internal combustion engine and compound parabolic concentrator PVT collector.

Drying is the process of conveying heat energy from heat source to a material and transmitting moisture from its surface to the environment. Typically, solar dryer mainly consists of air heater, drying chamber, and airflow system. Active or passive, direct, indirect or mixed solar dryers are the main categories of the solar dryers. Also, there are three main types of drying chambers: tent, cabinet, and greenhouse. Tiwari et al. (2009) proposed a design of mixed type PVT solar dryer as shown in Fig. 14. The dryer mainly consists of FPC, drying chamber, PV cells, and DC fan. The authors assessed the performance of the designed solar dryer by analyzing the thermal energy, exergy, and electrical energy gain of the system. The analysis was conducted under no load conditions for four distinct weather conditions and five different cities (Jodhpur, Bangalore, New Delhi, Srinagar, and Mumbai). The results showed that the best place to install PVT mixed solar dryer is Jodhpur.

In addition, Tiwari et al. (2018b) conducted a thermal modeling on PVT greenhouse solar drying system which is composed of PVT air collector and greenhouse solar dryer (Fig. 15). It was found that at 0.01 kg/s mass flow rate of air the electrical, thermal and overall efficiency are 11.26%, 26.68% and 56.3% respectively.

Fterich et al. (2018) developed an active mixed PVT solar dryer which is shown in Fig. 16. Experimental testing was performed to examine the system performance. The results revealed that the

Table 5
Theoretical and experimental works on PVT systems' applications.

St. Nb	Reference(Author/s, Year)	Application	Operating conditions	Efficiency Thermal	Electrical	Total	Type of study
	Hung et al. (2001)	Water heating	• Total radiation between 200 and 900 W/m2 • Ambient temperature between 25 and 35 °C	44.5%	9.1%	53.5%	• Experimental
	Zondag et al. (2003)	Water heating	• Ambient temperature of 20 °C • Sky temperature of 4 °C • Incident angle 45° • Wind velocity of 1 m/s • Total radiation of 800 W/m2	52%-58%	8.1%-9.7%	–	• Theoretical • Experimental
	Mittelman et al. (2007)	Space cooling	–	59%	20%	–	• Theoretical
	Barnwal and Tiwari (2008)	Drying	• Ambient temperature between 31 and 40 °C • Total solar intensity 148 and 398 W/m2	28.4%	10.8%	–	• Theoretical • Experimental
	Kumar and Tiwari (2008)	Desalination	• Wind velocity between 1 and 10.2 m/s • Ambient temperature between 19.6 and 28.2 °C	15%-20%	–	–	• Theoretical • Experimental
	Solanki et al. (2009)	Space heating	• Solar radiation of 600 W/m2 • Inlet fluid temperature of 38 °C	42.0%	8.4%	51%	• Theoretical • Experimental
	Ibrahim et al. (2009b)	Water heating	• Solar radiation of 800 W/m2 • Wind velocity of 1 m/s • Ambient temperature of 26 °C	50.12%	11.98%	68%	• Theoretical
	Adelaja et al. (2009)	Drying	• Ambient temperature of 28.3 °C	83.2%	–	–	• Experimental
	Xu et al. (2011)	Water heating	• Solar radiation between 500 and 900 W/m2 • Wind velocity of 0.75 m/s • Ambient temperature between 34.2 and 38.5 °C	–	17.5%	–	• Experimental
	Dupeyrat et al. (2011)	Water heating	• Total radiation of 960 W/m2 • Ambient temperature between 30 and 33 °C • Wind velocity of 3 m/s	79%	8.8%	–	• Theoretical • Experimental
	Eicker and Dalibard (2011)	Space cooling	• Total irradiance up to 400 W/m2 • Ambient temperature between 5 and 15 °C • Ambient temperature between −15 and −3°C	–	13%	–	• Theoretical • Experimental
	Bambrook and Sproul (2012)	Space heating	• Solar radiation of 800 W/m2 • Ambient temperature of 17 °C	28.0%-55.0%	10.6%-12.2%	–	• Experimental
	Aktas et al. (2012)	Drying	–	49.33%	–	–	• Experimental
	Bennamoun (2013)	Drying	• Solar radiation between 600 and 1000 W/m^2	58%	–	70%	• Theoretical
	Kim et al. (2014)	• Water heating • Space heating	–	30%	17%	47%	• Experimental
	Kiran and Devadiga (2014)	Water heating	• Inlet water temperature between 28.6 and 32.9 °C • Total radiation between 473.1 and 951.6 W/m^2 • Ambient temperature of 25 °C • Sky temperature of 19 °C	50.81%	8.16%	58.9%	• Experimental
	Calise et al. (2014)	Desalination	–	50%	20%	–	• Theoretical
	Saeedi et al. (2015)	Desalination	• Total radiation between 80 and 890 W/m^2 • Ambient temperature between 27 and 47 °C • Wind velocity of 1 m/s	17%	–	–	• Theoretical • Experimental
	Singh et al. (2016)	Desalination	• Wind velocity between 0 and 1 m/s • Solar radiation up to 500 W/m^2 • Ambient temperature between 9 and 24.5 °C	69.06%	–	–	• Theoretical • Experimental
	Alobaid et al. (2017)	Space cooling	–	–	10–35%	–	• Theoretical
	Ong et al. (2017)	Water heating	• Ambient temperature between 29 and 39 °C • Solar radiation between 20 and 880 W/m^2 • Inlet water temperature between 30 and 35 °C	–	0.16%	–	• Experimental
	Singh (2017)	Desalination	• Ambient temperature between 20 and 35 °C • Wind velocity between 1.83 and 4.11 m/s	–	12%	28%	• Theoretical
	Fan et al. (2018)	Space heating	• Wind velocity between 0 and 8 m/s • Sky temperature between 10 and 22 °C • Beam radiation up to 810 W/m^2	48.8%-56.9%	–	–	• Theoretical • Experimental
	Naroei et al. (2018)	Desalination	• Ambient temperature between 20 and 30 °C • Wind velocity between 0.5 and 3.7 m/s • Solar radiation between 150 and 800 W/m^2	19%	9.5%	33%	• Theoretical • Experimental
	Eltawil et al. (2018)	Drying	• Ambient temperature between 29 and 39 °C • Wind velocity between 0.1 and 1.3 m/s	41.2%-52.2%	–	14.05%-19.07%	• Theoretical • Experimental
	Tiwari et al. (2018b)	Drying	• Ambient temperature between 35 and 43 °C • Total radiation between 100 and 900 W/m^2 • Incident angle of 30°	11.26%	26.68%	56.3%	• Theoretical
	Maatallah et al. (2019)	Water heating	• Ambient temperature between 32 and 38 °C • Total radiation between 390 and 560 W/m^2	26.87%	13.72%	40.59%	• Experimental

(*continued on next page*)

Table 5 (*continued*)

St. Nb	Reference(Author/s, Year)	Application	Operating conditions	Efficiency Thermal	Electrical	Total	Type of study
	Misha et al. (2019)	Water heating	• Inlet water temperature of 26 °C • Solar irradiance of 600, 800 and 1000 W/m^2 • Wind velocity of 1 m/s • Ambient temperature of 30 °C	59.6%	11.71%	–	• Theoretical • Experimental
	Fu et al. (2019)	Water heating	• Ambient temperature between 25 and 35 °C • Solar radiation between 100 and 900 W/m^2	45.46%-60.11%	–	–	• Experimental
	Haiping et al. (2019)	Desalination	• Wind velocity of 1.8 m/s • Ambient temperature between 31.72 and 40.12 °C	–	13.28%	–	• Experimental

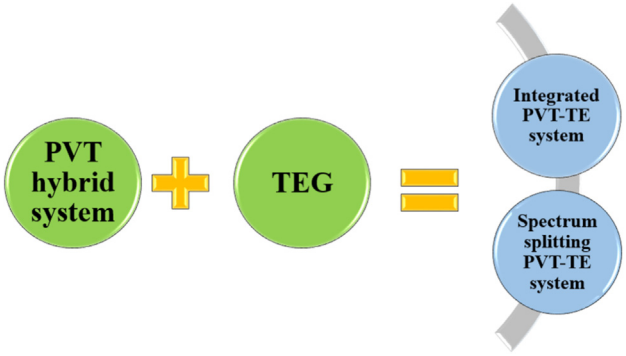


Fig. 18. Types of PVT-TE system.

combination of PVT air collector enhanced the drying rate of the solar dryer.

Tiwari et al. (2016) suggested PVT greenhouse solar dryer and assessed several parameters for the climatic condition of the Indian Institute of Technology. A thermal model is developed on the studied system based on several parameters such as crop, greenhouse, and solar cell temperatures. It was revealed that the experimental and theoretical global thermal energies are 2.03 kWh and 1.92 kWh respectively.

Solar desalination is a technology which utilizes SE to desalinate water by providing heat from sunlight for evaporative desalination process. Rafiei et al. (2019) investigated and analyzed the effect of the shape of the receiver on the performance of solar PVT desalination system. The system consists of parabolic dish collector, PV panel and humidification-dehumidification desalination unit as shown in Fig. 17. The three examined shapes of the receiver are cylindrical, cubical and hemispherical. It was obtained that the hemispherical cavity receiver produces the highest amount of freshwater and the lowest gain output ratio.

Also, Elsafi (2017) conducted mathematical model to demonstrate the performance of air-heated humidification-dehumidification desalination system combined with concentrated PVT collector. The cost of the system components was estimated by performing exergy-costing method. The results revealed that for a certain location having annual solar irradiance of 1.88 MWh/year, the system is capable to generate 960 kWh electricity and produce 12 m^3 of freshwater. Also, it was obtained that the unit cost for generating electricity is 0.289 $/kWh and for producing fresh water is 0.01 $/L. Mittelman et al. (2009) suggested a system which combines multi-effect evaporative desalination plant with concentrating PVT collector. The system generates electricity and simultaneously absorbs waste heat from PVM for desalination process. The authors conducted an elaborated simulation to calculate the electricity and water produced per year. Taking into consideration various economic cases, the water cost was computed and compared to that of traditional solar desalination plants. It was observed that the suggested system is more beneficial and in some scenarios it is more cost-effective compared to traditional system. Palomino-resendiz et al. (2018) proposed, constructed and tested a solar desalination system amalgamated with Fresnel lens PVT collector. It was concluded that the use of sun tracking system increases the concentrator efficiency leading to the rise in the thermal energy used and thus accelerating the desalination process even in the presence of environmental disturbances. Sotehi et al. (2016) implemented a prototype of solar energy system which covers the annual air conditioning and heating, domestic hot water and lightening in addition to driving several home appliances and solar still. The authors established a numerical simulation on TRNSYS software to simulate the performance of solar still. It was concluded that solar PVT system can cover the required thermal and electrical energy needed.

Table 5 elucidates the theoretical and experimental researches that were established on solar PVT systems' applications.

Analysis of Table 5 yields to the following concluding points:

- PVT systems are mostly utilized for heating water in comparison to the other applications.
- PVT systems are suitable for applications that require thermal

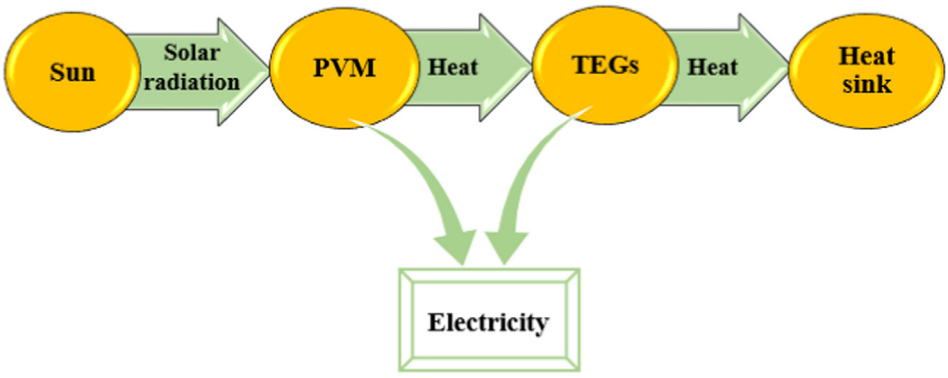


Fig. 19. Integrated PVT-TE system.

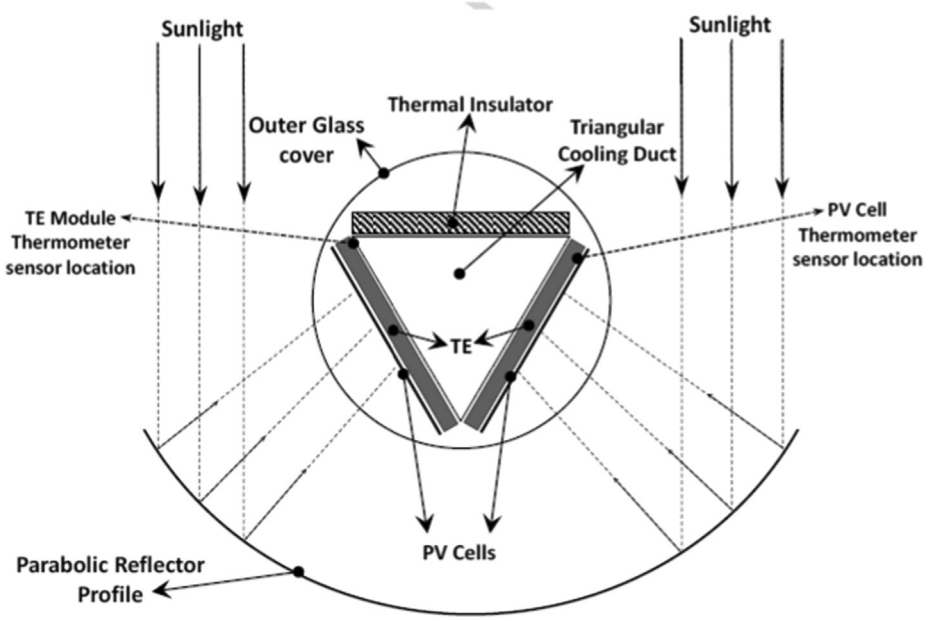


Fig. 20. Parabolic trough PVT-TE suggested by (Mohsenzadeh et al., 2017).

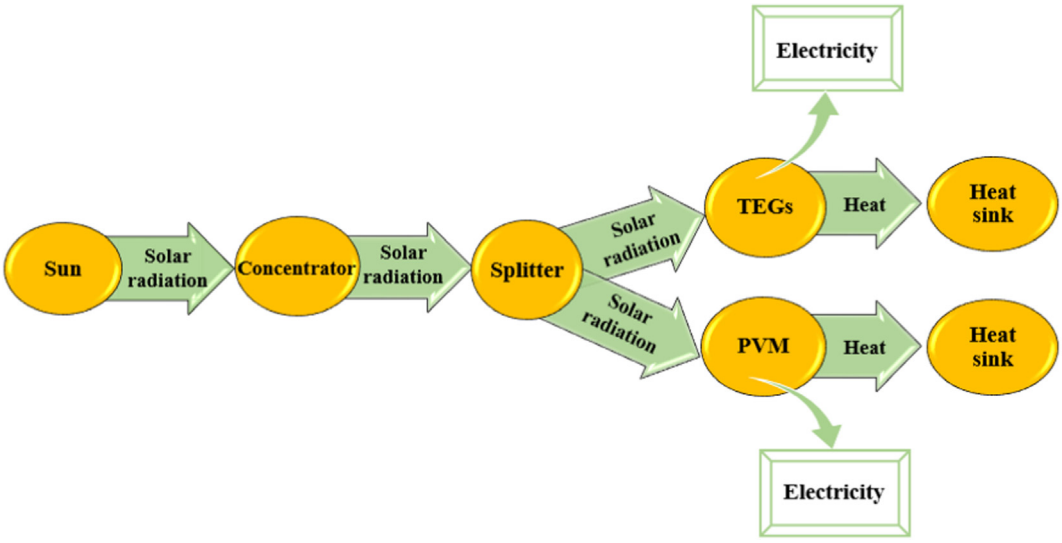


Fig. 21. Spectrum splitting PVT-TE system.

energy more than electrical energy.

- The outputs and performances of the PVT systems in all applications are better than that of thermal systems and PV systems working alone.
- In most applications, PVT systems are capable to cover the thermal and electrical energy required.

5. PVT system using thermoelectric generators

The integration of ThermoElectric Generators (TEGs) into solar PVT system has been recently adopted for the sake of optimizing the use of SE. The role of TEG is to transform thermal energy into electrical energy by Seebeck effect. The principle of this effect states that the existence of temperature difference on the junction of two different metals leads to production of an induced voltage. TEG can utilize the excess heat produced by PV cell and generate electricity. Thus, the electric conversion efficiency of the system can increase. Such technology is more preferred to be used in concentrated systems since it generates higher electricity at higher operating temperatures.

The main advantages of TEGs is that they operate with no moving parts and they have small sizes. Also, TEG integration rises the energy produced per area since TEG does not require additional place. In addition, the amount of generated power is extremely higher than that produced by standalone PV system, solar TEG system or PVT system.

The two main technologies that are adopted to amalgamate TEG with PVT system are: integrated PVT-TE system and spectrum splitting PVT-TE system (Fig. 18).

Integrated PVT-TE system mainly consists of solar thermal collector, PVM and TE module. In most systems the TEG is fitted on the back face of the PV cell to exploit heat from it and generate electricity (Fig. 19).

Mohsenzadeh et al. (2017) suggested a new layout for parabolic trough PVT collector and performed an experimental investigation to analyze the global system performance. The receiver of the collector includes a triangular channel covered with PV cell and TEGs as shown in Fig. 20. It was observed that the daily average electrical and thermal efficiencies achieved by the hybrid system are 4.83% and 46.16%

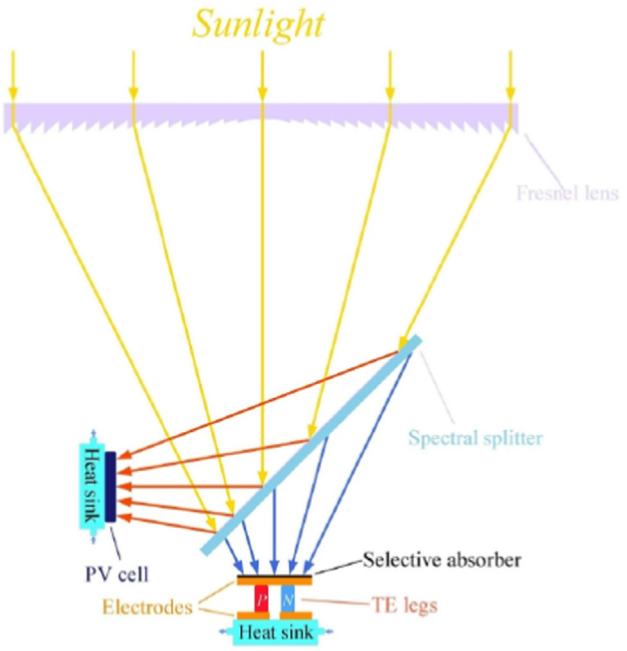


Fig. 22. Concentrated spectrum splitting PVT-TE system studied by (Yin et al., 2018).

respectively.

Yin et al. (2019) suggested a new hybrid cogeneration concentrating PVT thermoelectric system. The authors compared the operating temperature, energy efficiency and exergy efficiency of the proposed system and the existing systems in order to expose the notability of the new system. Also the influence of thermoelectric structure and controlled temperature on the system performance were demonstrated by the study as well.

Kohan et al. (2018) established three dimensional numerical models for PVM and TEG to analyze the performance of PVT system with TEGs located at the PV layer back part. Results showed that the power generated by the hybrid system is higher than that generated by traditional PV and the TEG are not efficient in cooling the PVM.

Shadmehri et al. (2018) developed a detailed three-dimensional computational model and numerical simulations on a novel concentrating PVT system coupled with TEG. The impact of temperature and cooling on the PVM performance is discussed. Also, the influence of varying the reflector aperture width and the absorber apex angle on distribution of solar energy flux and output power of PVM were investigated.

Nazri et al. (2019) conducted experimental and theoretical study to investigate the performance of PVT-TE air collector. Air outlet temperature and PV temperature are computed using a steady state thermal model. The model was also used to evaluate the influence of mass flow rate on exergy efficiency. In addition, Syakirah et al. (2018) performed a theoretical analysis to investigate the performance of PVT-TE collector by suggesting and solving a mathematical model based on iterative procedure.

Li et al. (2018) presented a review on PVT-TE system including the operation and feasibility of the system. Also, the review comprises the main types, characteristics and performance of PVT-TE systems. Moreover, the methods of optimizing and developing such system are exposed in the review.

Chávez-Urbiola et al. (2012) demonstrated the potential of integrating TEG into hybrid PVT system. The authors examined four systems of different configurations. It was detected that the thermal stability and electrical efficiency were optimized when TEG are combined to the PVT system.

Besides, Arturo et al. (2013) carried out experimental investigation on concentrating PVT system with six series connected thermoelectric generators. It was found that the system produced 200 W thermal energy and 20 W electrical energy.

Jadin et al. (2017) developed a hybrid PVT-TE system to increase the power generation efficiency of the system. It was observed that the maximum power output was 99.27 W which was higher than that generated in case of individual system.

Dimri et al. (2018) investigated the effect of working fluid and type of PVM on the performance of PVT-TE collector. The studied working fluids are water and air, while the considered PVM are opaque aluminum base, and semitransparent. It was noticed that the electrical and thermal efficiencies of opaque water system had higher thermal and electrical efficiencies compared to opaque air system. Also, it was found that the daily gained electrical energy, thermal energy and exergy were the highest for aluminum base PVT-TE water collector. Moreover, Dimri et al. (2017) developed thermal modeling for semitransparent PVT-TE collector by utilizing energy balance. The authors compared the electrical efficiency of the system with that of semitransparent PV and PV-TE collectors. They concluded that the semitransparent PVT-TE

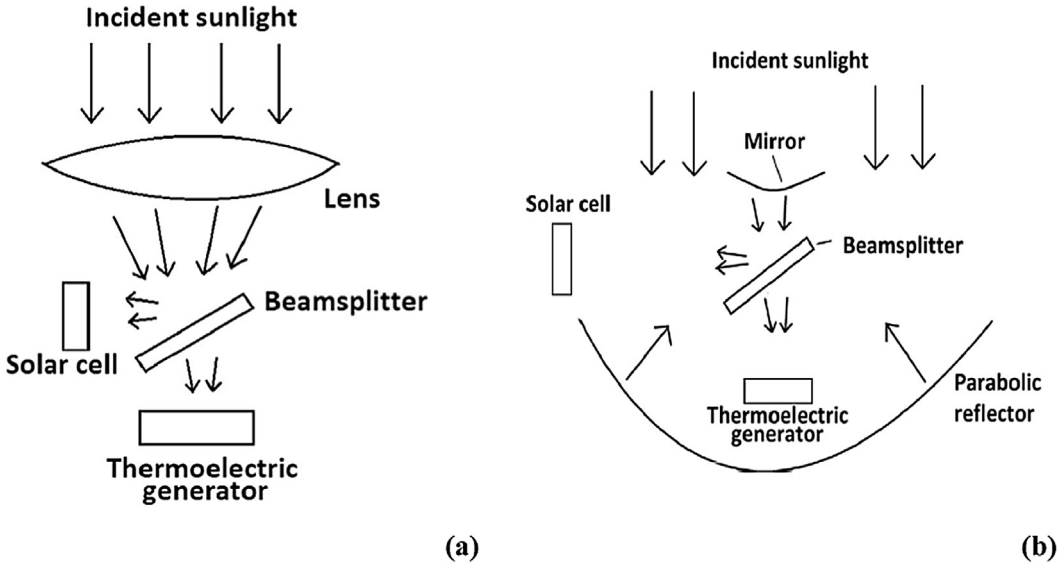


Fig. 23. Spectrum splitting PVT-TE system studied by (Skjølstrup and Søndergaard, 2016).

Table 6
Researches performed on PVT-TE hybrid systems.

St. Nb	Reference(Author/s, Year)	PVT-TE system	Operating conditions	System performance	Type of study
	Sark (2011)	Integrated	• Ambient temperature of 25 °C • PVM temperature of 80 °C • Solar radiation of 1000 W/m^2	• Maximum TE efficiency of 5% • PV electrical efficiency of 14.03%	• Theoretical
	Ju et al. (2012)	Spectrum splitting	• Ambient temperature of 25 °C • Maximum TEG temperature of 527 °C • Maximum PVM temperature of 227 °C	• PV-TE hybrid system efficiency ranges between 26.62% and 27.49% • TEG generates 10% of the overall output power of the system	• Theoretical
	Xu et al. (2015)	Integrated	–	• TE electrical efficiency of 2.2% • PVT-TE electrical efficiency of 18%	• Theoretical
	Attivissimo et al. (2015)	Integrated	• PVM temperature between 50 and 60 °C	• Electrical efficiency ranges between 13% and 16% • Total efficiency ranges between 45% and 70%	• Theoretical
	Cui et al. (2016)	Integrated	• Ambient temperature of 27 °C • PV cell temperature between 21 and 35 °C • Water inlet temperature of 20 °C • Total solar irradiance of 1000 W/m^2	• Experimental TE efficiency between 2.1 and 2.58% while the simulated result between 2.1 and 2.8% • Performance of PV-PCM-TE system is better than that of PV system and PV-TE system.	• Theoretical
	Zhu et al. (2016)	Integrated	• Ambient temperature of 27 °C • Solar irradiation of 1000 W/m^2	• Peak PV-TE system efficiency of 23%	• Experimental • Theoretical
	Cotfas et al. (2016)	Integrated	• Solar irradiance between 720 and 1080 W/m^2	• The PV cell efficiency is 18.65% for mono-crystalline silicon and 16.43% for poly-crystalline silicon at solar irradiance of 1000 W/m^2	• Theoretical
	Kossyvakis et al. (2016)	Integrated	• Cold side temperature of TEG of 20 °C • Solar irradiation of 1000 W/m^2	• Power output enhancement using poly-crystalline silicon reaches 22.5%	• Experimental
	Li et al. (2016)	Integrated	• Solar radiation between 100 and 1000 W/m^2 • Wind speed between 1 and 8 m/s • Ambient temperature between 15 and 40 °C	• Electrical efficiency is 18% higher than that of PV system • TE efficiency of 2% • Maximum PV-TE system efficiency of 11.3%	• Theoretical
	Skjølstrup and Søndergaard (2016)	Spectrum splitting	–	• Efficiency of hybrid system is higher than that of PV system by 21.4%	• Theoretical
	Ong et al. (2017)	Integrated	• Ambient temperature between 29 and 39 °C • Solar radiation between 20 and 880 W/m^2 • Inlet water temperature between 30 and 35 °C	• Total electrical efficiency of 0.16%	• Experimental
	Li et al. (2017)	Integrated	• Ambient temperature of 27 °C • Water inlet temperature of 27 °C • Solar radiation of 1000 W/m^2	• Maximum module TE efficiency is 6.1% • Peak PV-TE electrical efficiency of 21.6%	• Theoretical
	Soltani et al. (2018)	Integrated	• Ambient temperature of 36.2 °C • Water inlet temperature of 28 °C • Solar radiation of 900 W/m^2	• Maximum thermal efficiency of 57% • Maximum TE efficiency of 5.6%	• Theoretical
	Sharaf and Orhan (2018)	Integrated	• Ambient temperature between 19.9 and 34.9 °C • Inlet water temperature equal to ambient temperature	• Thermal efficiency of 78.9% for parallel configuration and 76.3% for series configuration	• Theoretical
	Yang et al. (2018)	Spectrum splitting	–	• PVT-TE system efficiency is 39.5% and 40.2% at concentration ratio of 30 and 100 respectively	• Theoretical
	Yin et al. (2018)	Spectrum splitting	• Ambient temperature of 27 °C • Cooling fluid temperature of 20.13 °C • TEG hot side temperature of 527 °C	• TE efficiency of 13.9% • PVT-TE maximum efficiency of 30%	• Theoretical
	Shittu et al. (2019)	Integrated	• Ambient temperature between 15 and 55 °C • Wind speed between 0 and 7 m/s	• PVT-TE system efficiency is higher than that of PV-TE system by 1.47% and PV system by 61.01% at concentration ratio of 6	• Theoretical
	Tiwari et al. (2020)	Integrated	• Ambient temperature between 10 and 38 °C • Solar intensity up to 1000 W/m^2	• Hourly electrical gain ranges between 3.96 and 14.38 W • Hourly rate of thermal energy gain ranges between 41.14 and 148.72 W	• Experimental • Theoretical
	Riahia et al. (2020)	Integrated	• 760 and 925 W/m^2 • Inlet fluid temperature between 28.8 and 35.8 °C • Ambient temperature between 36.5 and 33 °C	• Daily electrical efficiency of PVT-TE is higher than that of PVT system by 7.46%	• Experimental • Theoretical
	Shittu et al. (2020)	Integrated	• Solar irradiance of 1000 W/m^2	• Maximum PVT-TE electric efficiency of 15%	• Theoretical

collector had electrical efficiency greater than semitransparent PV-TE collector by 4.723% and semitransparent PV collector by 7.266%.

Temperature has an opposing influence on the performance of TEG and PV. Higher temperature difference raises the efficiency of TEG but at the same time increases the temperature of PV cell leading to alleviate the PV efficiency. Due to this, spectrum splitting system has been suggested which reflects the appropriate solar radiation to the PVM and the remaining out of range radiation to the TE module (Fig. 21). Spectrum splitting PVT-TE system is generally composed of solar concentrator, PVM, TEG and spectrum splitter.

Compared to integrated PVT-TE system, rare are the studies that are conducted on spectrum splitting PVT-TE system. Ju et al. (2012) developed a numerical modeling and optimization analysis of spectrum splitting PV-TE system. The authors conducted a simulation model taking into account the solar concentration and various heat loss rates. They also carried out analysis to estimate the electrical and thermal performance for distinct conditions. It was obtained that the efficiency of the studied system ranges between 26.62% and 27.49%.

Yang et al. (2018) conducted analytical expression of the system efficiency by utilizing an updated model of solar spectrum PV-TE system. They optimized the cell voltage, cutoff energy of the system and dimensionless current of the TEG. Also, they studied the impact of the ratio of the collector area to PV cell area on the performance of spectrum splitting system. The results revealed that for concentration ratio of 30 and 100, the maximum system efficiency achieved is 39.5% and 40.2% respectively where it increased by about 2.67% and 2.19% in comparison to PV system.

Yin et al. (2018) suggested a new design method to optimize the solar energy distribution of concentrated spectrum splitting PV-TE hybrid system. The system comprises concentrator (Fresnel lens), PV cell, TE module and cooling system (Fig. 22). A model was conducted in order to investigate the performance of the system and reach the optimum design. The results showed that the maximum system efficiency attained is 30% and the TEG efficiency is 13.9%.

Skjølstrup and Søndergaard (2016) designed and optimized a thin-film multilayer spectrum beam splitter for PV-TE hybrid system. The authors considered two configurations where the first one uses lens and the second one uses reflector (Fig. 23). It was obtained that the studied system efficiency is greater than the efficiency of PVM by 21.4%.

Table 6 summarizes the latest experimental and theoretical researches performed on PVT-TE hybrid systems.

Several points can be concluded after analysis of the outputs of the previous studies:

- The most commonly used type of PVT-TE system is the integrated type where it is simple in construction compared to the spectrum splitting.
- Utilizing TEG serves in increasing the power output and the electrical efficiency of PVT system.
- PVT-TE systems exhibit better performances than PVT systems.

6. Closure and discussion

The present work comprises a review on hybrid solar system which amalgamates solar PV with solar thermal collectors. The hybrid PVT system simultaneously provides thermal and electrical energy. Hence, the main aim of this cogeneration system is to maximize the avail of solar energy.

The review includes general background which states the history, concept and advantages of PVT systems. Also, the main classifications of PVT system are presented where it is categorized in this study according to the type of PV cell and solar collector, working fluid, glazing and absorber. Besides, solar applications derived from PVT hybrid system are described in this work. The main application of such systems is producing electricity in addition to other purposes such as drying, heating water, heating and cooling space and desalination. In addition, the review includes the description of hybrid PVT system with thermoelectric generators in which the TEG can be integrated with PV cell or they can operate separately using spectral beam splitter.

The main outputs of the review are:

- Crystalline silicon and InGaP/GaAs/Ge triple junction solar cells most utilized PV cells in PVT systems, and high electrical efficiency is obtained when utilizing InGaP/GaAs/Ge triple junction PV cell compared to crystalline silicon.
- Thermal efficiency of PVT system is slightly affected by the type of PV cell.
- The selection of the solar collector depends on the type of application where each one requires certain range of outlet temperature.
- Concentrated type of STC; mainly parabolic trough and linear Fresnel are the most utilized in PVT systems. The electrical efficiency is negligibly affected by the type of the STC.
- Water based system is limited by the operating conditions due to freezing at low operating temperatures, but it is preferred to use water since it has high thermal characteristics compared to air. Water based PVT system has higher electrical and thermal efficiencies compared to air based PVT system.
- Glazing is suitable for increasing the thermal efficiency but it leads to mitigate the electrical efficiency of the system. Unglazed PVT system is more favorable if high electrical output is required while glazing is more preferred to be used in PVT systems in case high thermal or overall system efficiency is required. However, single glazing is more preferable than double glazing.
- The type of absorber plays a great role in exploiting solar energy where aluminum and copper are the commonly used materials in the solar thermal absorbers.
- PVT systems are suitable for applications that require thermal energy more than electrical energy. Water heating is relatively the most application that exploits PVT system.
- In all applications, the performance PVT systems is better than that of thermal systems and PV systems working alone. PVT systems can cover the thermal and electrical energy required by most applications.
- The most commonly utilized type is the integrated type of PVT-TE systems in which TEG are integrated at the PV cell back face. Using TEG serves in raising the electrical efficiency of the system and thus the overall efficiency.
- PVT-TE systems have better performance than PVT systems.

It is predicted that PVT systems will be capable to suffice the need for energy consumption in the near future. Also, these systems provide energy needed in a clean and environment friendly ways. This explains the significance of such systems and thus more researches and studies must be conducted in order to enhance and optimize the system for better efficiency and less costs.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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